

Relation Algebra

§1. Notation

Traditionally Adjunction $F \dashv G$ is defined as $F(x) \leq y$ iff $x \leq G(y)$, and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \leq Y$ iff $X \leq YG$

• write F as $/$, G as $\backslash$

• render $a \leq b$ as $\boxed{\begin{smallmatrix} a \\ b \end{smallmatrix}}$

• **rewrite** $X \leq XF$ as $1 \leq F$

(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj

$$\begin{array}{c} \boxed{\frac{X/}{X/}} \xRightarrow{\text{adj}} \boxed{\frac{X}{X\backslash}} \xRightarrow{\text{rewrite}} \boxed{\frac{*}{\bigwedge}} \begin{array}{l} \xRightarrow{\text{put } \backslash \text{ on left}} \boxed{\frac{\backslash}{\bigwedge}} \\ \xRightarrow{\text{put } / \text{ on right}} \boxed{\frac{/}{\bigvee}} \end{array} \\ \text{unit} \\ \boxed{\frac{Y\backslash}{Y\backslash}} \xRightarrow{\text{adj}} \boxed{\frac{Y\bigvee}{Y}} \xRightarrow{\text{rewrite}} \boxed{\frac{\bigvee}{*}} \begin{array}{l} \xRightarrow{\text{put } / \text{ on left}} \boxed{\frac{\bigvee}{/}} \\ \xRightarrow{\text{put } \backslash \text{ on right}} \boxed{\frac{\bigwedge}{\backslash}} \end{array} \\ \text{counit} \end{array}$$

now we have $\bigvee = /$ and $\bigwedge = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

$F \dashv G$	F 's type	monad FG	$1 \in FG$	$GF \in 1$	F preserves u	G preserves n	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleright$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \in \triangleleft \triangleright$	$\triangleright \triangleleft \in 1$	$(RuS) \triangleleft = R \triangleleft uS \triangleleft$	$(RnS) \triangleright = R \triangleright nS \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow I$	$\multimap \multimap = T$	$1 \in \multimap \multimap$	$\multimap \multimap \in 1$	$(RuS) \multimap = R \multimap uS \multimap$	$(RnS) \multimap = R \multimap nS \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$\circ \circ = 1$	$R = R \circ \circ$	$R = R \circ \circ$	$(RuS) \circ = R \circ uS \circ$	$(RnS) \circ = R \circ nS \circ$	$R \circ \circ \circ = R \circ$	$R \circ \circ \circ = R \circ$
$\multimap \triangleleft \dashv \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \multimap \dashv \multimap \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap \triangleleft) (\triangleright \multimap \otimes 1) = 1$	$(\multimap \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv n$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \mapsto RnR = R$	$R \in RnR$	$RnS \in R$	$\Delta(RuS) = \Delta Ru \Delta S$	$(RnT) n (SnU) = (RnS) n (TnU)$	$RnR = R$	$RnR = R$
$u \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (RuS, RuS)$	$R \in RuS$	$RuR \in R$	$(RuT) u (SuU) = (RuS) u (TuU)$	$\Delta(RnS) = \Delta Rn \Delta S$	$RuR = R$	$RuR = R$
$\perp \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$	—	—	$\perp \in R$	—	—	—	—
$\cdot S \dashv /S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \in (RS)/S$	$(T/S) S \in T$	$(RuT) S = RSuTS$	$(RnT)/S = R/SnT/S$	$((RS)/S) S = RS$	$((T/S) S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \in S \setminus (SR)$	$S(S \setminus T) \in T$	$S(RuT) = SRuST$	$S \setminus (RnT) = S \setminus RnS \setminus T$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$Rn \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (XnR)$	$X \in R \Rightarrow (XnR)$	$Rn(R \Rightarrow Y) \in Y$	$Rn(XuY) = (RnX) u (RnY)$	$R \Rightarrow (XnY) = (R \Rightarrow X) n (R \Rightarrow Y)$	$Rn(R \Rightarrow (XnR)) = XnR$	$R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \in (\mathcal{D}R)T$	$\mathcal{D}(AT) \in A$	$\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$	$(AnB)T = ATnBT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \in T(\mathcal{R}R)$	$\mathcal{R}(TA) \in A$	$\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$	$T(AnB) = TAnTB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	ff°	$1 \in ff^\circ$	$f^\circ f \in 1$	$(RuS)f = RfuSf$	$(RnS)f^\circ = Rf^\circ nSf^\circ$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	ff°	$1 \in ff^\circ$	$f^\circ f \in 1$	$f^\circ(XuY) = f^\circ Xu f^\circ Y$	$f(XnY) = fXn fY$	$ff^\circ f = f$	$f^\circ ff^\circ = f^\circ$
$i \dashv E$	$\text{Map} \hookrightarrow \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow E \quad A$	$\exists : E \rightarrow B \quad B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, E \rightarrow B) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R}{\exists} \exists = R$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = f \exists$	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$
$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, E \rightarrow B)$	1	$\frac{R}{\exists} \exists = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$	$\frac{f \exists}{\exists} \exists = f \exists$

§1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
\boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} (f^\circ X) T \\ R \end{array}} \\
\text{f a map} \\
\\
\xLeftrightarrow[\text{associativity}] \boxed{\begin{array}{c} f^\circ(X T) \\ R \end{array}} \xLeftrightarrow[f^\circ \cdot \dashv f \cdot] \boxed{\begin{array}{c} X T \\ f R \end{array}} \xLeftrightarrow[\cdot T \dashv /T] \boxed{\begin{array}{c} X \\ (f R)/T \end{array}} \\
\\
\xRightarrow[\text{indirect equality}] \boxed{\begin{array}{c} f(R/T) \\ (f R)/T \end{array}}
\end{array}$$

	$\cdot T$	$T \cdot$	$\cdot g$	$g^\circ \cdot$	Tn	$^\circ$	$\circ \triangleleft$	Δ	(1.2b)
$\cdot S$	$R/(ST) = (R/T)/S$	$T \setminus (R/S) = (T \setminus R)/S$	$R/(Sg) = (Rg^\circ)/S$	$g(R/S) = (gR)/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(T_1 n T_2)/S = T_1/S n T_2/S$	
$S \cdot$	$S \setminus (R/T) = (S \setminus R)/T$	$(TS) \setminus R = S \setminus (T \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ/S^\circ$	$((S \otimes 1)R)^\wedge = S \ R^\wedge$	$S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$	
$\cdot f$	$R/(fT) = (R/T)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(fg)^\circ = g^\circ f^\circ$	—	—	$(fY)^\circ = Y^\circ f^\circ$	—	$(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$	
$f^\circ \cdot$	$f(R/T) = (fR)/T$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(fg)^\circ = g^\circ f^\circ$	—	$(fY)^\circ = Y^\circ f^\circ$	—	$f(T_1 n T_2) = fT_1 n fT_2$	
Rn	—	—	—	—	$(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$	
$^\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ/T^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(gY)^\circ = Y^\circ g^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$Y^\circ = Y$	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$	
$\circ \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = T \ R^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$ both bends: the $^\circ$ section's display	—	—	
u	$(X_1 u X_2)T = X_1 T u X_2 T$	$T(X_1 u X_2) = TX_1 u TX_2$	$(X_1 u X_2)g = X_1 g u X_2 g$	$g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$	$Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$	$(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$	—	—	
$\perp$	$\perp T = \perp$	$T \perp = \perp$	$\perp g = \perp$	$g^\circ \perp = \perp$	$Tn \perp = \perp$	$\perp^\circ = \perp$	—	—	

§1.3. Adjoint triples

Beyond $u \dashv \Delta \dashv n$, which §1.1 already carries as two rows, the table holds one adjoint triple with content: $\exists_f \dashv f^* \dashv \forall_f$ along a map f , once on each side of the composite. $f : A \rightarrow B$ throughout this subsection.

$$\begin{aligned} \cdot f &\dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot &\dashv f \cdot \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor: $\cdot S \dashv /S$ at $S := f^\circ$, and $S \cdot \dashv S \setminus$ at $S := f$.

The third link is not the first over again. For a map f , $/f$ collapses to $\cdot f^\circ$; being a map is exactly what that collapse spends $R/(fS) = (R/S)f^\circ$. — f° is not a map, so $/f^\circ$ does not collapse in turn, and the three operators stay distinct. On $\text{Rel}(\mathbb{I}, A) = E\ A$ they are the image triple:

operator	acts by	name	(1.3b)
$\cdot f$	$S \mapsto f[S]$	direct image, $\exists_f$	
$\cdot f^\circ$	$T \mapsto f^{-1}[T]$	inverse image, f^*	
$/f^\circ$	$S \mapsto \{b : f^{-1}(b) \subseteq S\}$	$\forall_f$	

§1.1's $\mathcal{D} \dashv \cdot T$ is this same chain along the projection $A \otimes B \rightarrow A$, read through $\text{Rel}(A, B) = E(A \otimes B)$: $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$, AT is the relation that ignores b altogether, and the third link is $R \mapsto 1 \cap R/T = \{(a, a) : \forall b. aRb\}$.

$$\mathcal{D} \dashv \cdot T \dashv 1n\cdot/T \quad (1.3c)$$

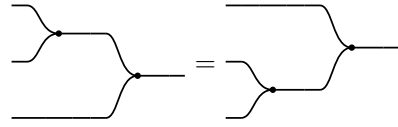
Three is where it stops, and one map breaks both ends. Take $A = \{a_1, a_2\}$ and $B = \{b\}$: $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$ while $f[\{a_1\} \cap \{a_2\}] = \emptyset$, so $\cdot f$ does not preserve meets and has no left adjoint, and the same two subsets give $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$ against $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$, so $/f^\circ$ does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — τf still reaches only $\text{im } f$, and $\forall_f \perp = B \setminus \text{im } f$ is still not $\perp$. The chain extends only when f° is a map as well, and then $/f^\circ = \cdot f^{\circ\circ} = \cdot f$ and it repeats forever. For an S with neither S nor S° a map there is no triple at all: $\cdot S \dashv /S$ is two links and stops.

The rows that chain forever say nothing by it: $^\circ$ is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and $^\circ \dashv \circ \dashv \circ$ has no end, and §1.1's other row of that kind, $\circ \triangleleft \dashv \triangleright \circ$, goes the same way.

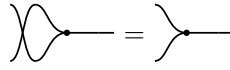
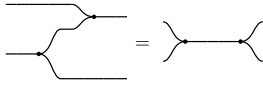
§2. Relations

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \circ$, and

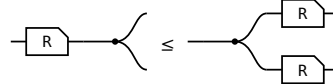
$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ : \mathbb{I} \rightarrow A)$ a commutative monoid with $\mathbf{C}^\circ \dashv \mathbf{C}$. We write $\mathbf{C}^\circ$ as $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow \mathbb{I})$



$\triangleright$ associative

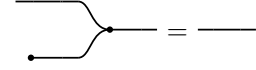


$\triangleright$ commutative

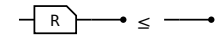


Frobenius, one half — the other is its $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$



$\circ$ is its unit

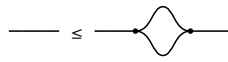


$R \circ \leq \circ$

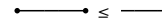
§2.1. $\forall$ object $\mathbf{A}$, $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



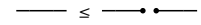
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft \triangleright$



$\circ \circ \leq 1$



$1 \leq \circ \circ$

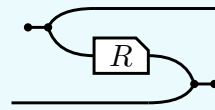
(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circ\text{--}\triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright\text{--}\circ)$$

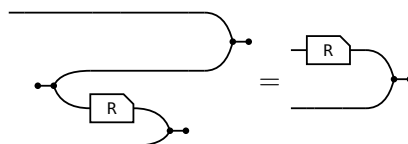
where $\circ\text{--}\triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright\text{--}\circ : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\text{--}\circ$ are $\triangleright^\circ$ and $\circ\text{--}$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

(3.1a)

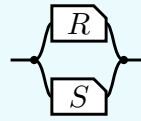


R° is DEFINED as the bending of $(R \otimes 1) \triangleright\text{--}\circ$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circ\text{--}\triangleleft$ bends the a strand down and the $\triangleright\text{--}\circ$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

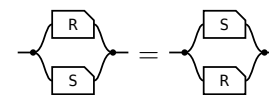
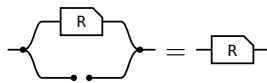
The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



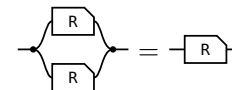
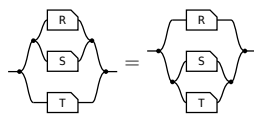
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).

(4b)



unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$ **commutative:** σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

And one law relating $\cap$ to composition, which is **not** an equation:

(4c)

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F (R \cap S) = F R \cap F S$.	

§5. Domain and range

The **domain** $\text{Dom } R \triangleq 1 \cap R R^\circ$ and the **range** $\text{Ran } R \triangleq \text{Dom } R^\circ$.



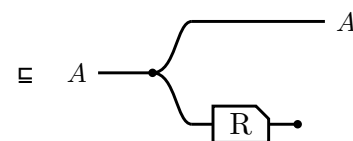
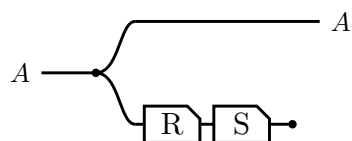
$\triangleright$ lands the return leg back on the value $\triangleleft$ handed out,
so $R^\circ \triangleright$ cuts to $\circ$ — Frobenius

Running R and throwing the result away leaves only the fact that R could fire, and $\text{Ran } R$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{Dom } R \triangleq 1 \cap R R^\circ$
$\text{Dom } R \subseteq A \iff R \subseteq A R, \text{ for } A \text{ coreflexive}$
$\text{Dom } (R S) \subseteq \text{Dom } R$
$\text{Dom } (R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Dom } R = 1 \iff 1 \subseteq R R^\circ$
$R \text{ simple} \iff R^\circ R \subseteq 1$
$R \text{ a map} \iff R \text{ entire and simple}$
$R, S \text{ entire} \implies R S \text{ entire} \text{ — likewise simple, likewise maps}$
$R S \text{ entire} \implies R \text{ entire}$

§5.1. Sliding the discard

$\text{Dom } (R S) \subseteq \text{Dom } R$, and a single glyph for Dom would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

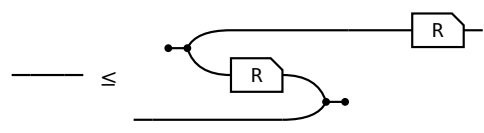
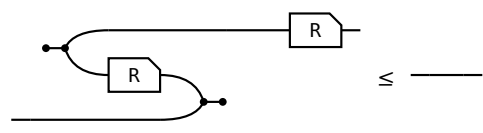
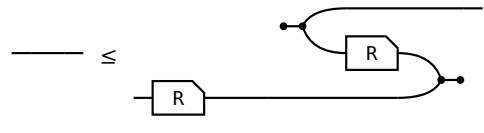
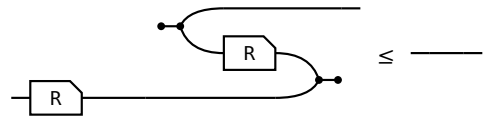


$S \circ \subseteq \circ$, the lax axiom for $\circ$ in the first section
— the discard slides back past S

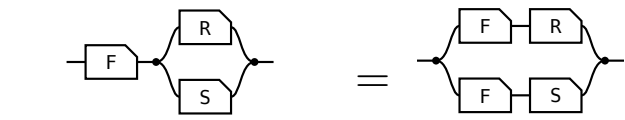
Equality is $S \text{ entire}$, which is the same picture read as $\text{Dom } R = 1 \iff R \text{ entire}$.

§6. Maps

Every arrow is lax for $\triangleleft$ and for $\multimap$ by axiom, and for $\triangleright$ and $\multimap$ by taking $\circ$ of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

 surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	 single valued $R^\circ R \sqsubseteq 1$
 entire $1 \sqsubseteq R R^\circ$. With single valued, a map.	 injective $R R^\circ \sqsubseteq 1$

(6a)

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

(6b)

§7. / is all of

(7a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/ compares images: $S(y) \subseteq R(x)$. $\setminus$ compares preimages: $S^\circ(x) \subseteq R^\circ(y)$.

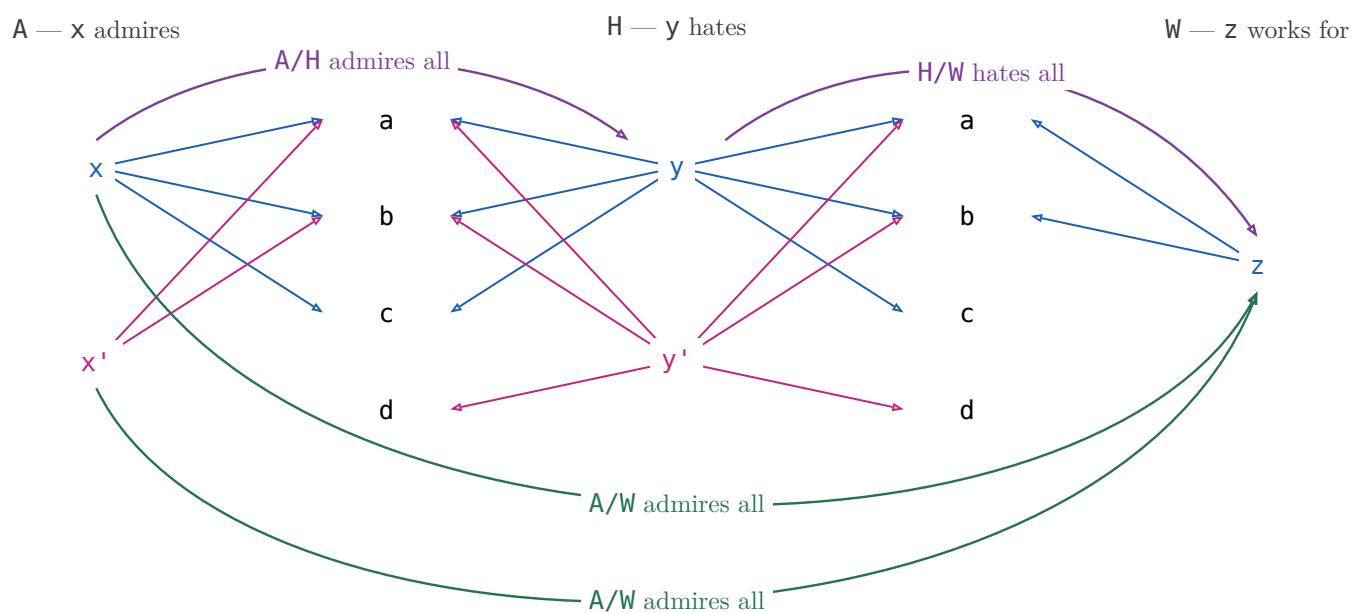
example: A admires, H hates, W works for

$x (A/H) y$ — x admires everyone y hates.

$x (H \setminus A) y$ — everyone who hates x admires y .

§7.1. $(R/S)(S/W) \sqsubseteq R/W$

(7.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(7.1b)

$(A/H)(H/W)$ is a path: $x \rightarrow y \rightarrow z$, and that is all of it. A/W also holds of x' , who admires everyone z works for — but x' does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of $(R/S)(S/W) \sqsubseteq R/W$.

$X \sqsubseteq R/S \iff X S \sqsubseteq R$ X is any X -to- Y pairing; one that only pairs X with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \sqsubseteq S \setminus R \iff S X \sqsubseteq R$ The mirror — divide on the left when X comes first.	
$(R/S) S \sqsubseteq R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \emptyset$: R/S is everyone, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f (R/S) = (f R)/S$ Rename x before or after dividing — the licence to write $f R / S$.	
$R/(f S) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S) (S/W) \sqsubseteq R/W$ Someone who admires all of a hate-set that already covers everyone Z works for admires those people too.	
$1 \sqsubseteq R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R) (R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \sqsubseteq R/R$ above it is an equality.	
$(R/R) R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§8. $\frac{R}{S}$

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$. In Rel x and y has the same image: $\forall p. (x R p \iff y S p)$

(8a)

$x (A/H) y$ — x admires everyone y hates

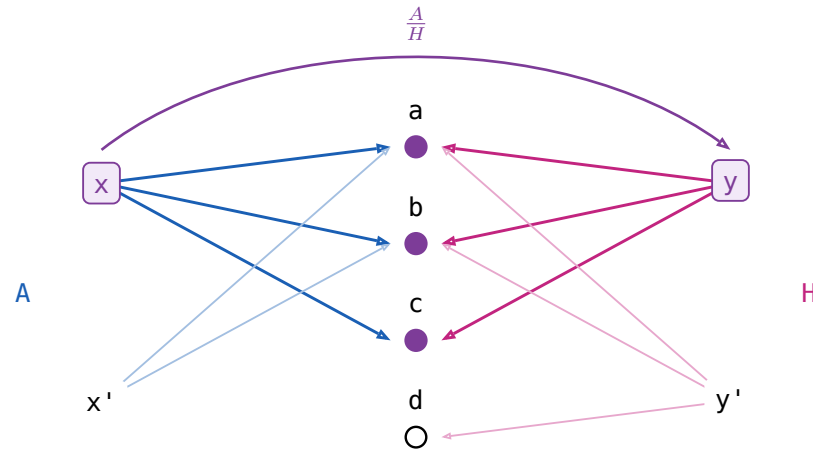
(8b)

$x ((H/A)^\circ) y$ — x admires only people y hates

$x ((A/H) \cap (H/A)^\circ) y$ — x admires only and all whom y hates

x admires exactly whom y hates: $/$ is **all of**, $\frac{R}{S}$ is **only all of**.

(8c)



law	the reading
$X \subseteq \frac{R}{S} \iff XS \subseteq R \text{ and } X^\circ R \subseteq S$	X may pair x with y only when x admires exactly whom y hates. Both halves must typecheck, so the operation is partial .
$(\frac{R}{S})^\circ = \frac{S}{R}$	Matching is symmetric.
$\frac{R}{S} \frac{S}{W} \subseteq \frac{R}{W}$	And transitive.
$\frac{R}{S} S \subseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$ x only admires whom y hates $\frac{R}{S} S = \text{Dom}(\frac{R}{S})R$
$\frac{R}{R} R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$ x and y admire the same people $y = x$ always qualifies ($1 \subseteq R\%R$ below)
$1 \subseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if x and y admires the same peoples.
$(\frac{R}{R})^2 = \frac{R}{R}$	So the relation admires the same people is an equivalence relation.
$X \subseteq \frac{R}{R} \iff XR \subseteq R$, for symmetric X	The largest symmetric arrow that leaves R alone.
$\frac{R}{1}$ is the simple part of R	The people who admire exactly one person and nobody else. It equals R only when R is simple, unlike $R/1 = R$.
$\text{Dom } \frac{R}{S} = 1 \cap (R/S)(S/R)$	Its domain is the domain of simplicity of R .

(8d)

§9. Freyd's Power allegories

(9a)

A **power allegory** is a division allegory with one unary operation on arrows, $\exists$ **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \frac{\exists_R}{\exists_R} &= \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R\square$ is R 's target, an identity arrow. For $R : A \rightarrow B$ write $\exists : E B \rightarrow B$, dropping the subscript.

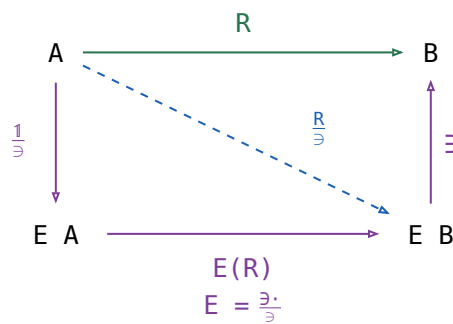
$$\epsilon \triangleq \exists^\circ : A \rightarrow E A$$

(9b)

law	the reading
1 $\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow E B$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\frac{R}{\exists}$ is a map	
5 $\frac{R}{\exists} \exists = R$	
6 $F \subseteq \frac{F \exists}{\exists}$, F simple	A partial choice of sets is inside the total one.
7 $\frac{1}{\exists}$, the singleton map , monic	The one-person set.
8 $\frac{\exists}{\exists} = \mathbb{1}$	Make the set of a set, then read it back one level down.
9 fusion: $\frac{f R}{\exists} = f \frac{R}{\exists}$, f a map	Naturality of the unit, $f \frac{1}{\exists} = \frac{1}{\exists} (E(f))$.
10 $\frac{f}{\exists} = f \frac{1}{\exists}$, f a map	Rename first or take singletons first — the fusion row above at $R = \mathbb{1}$.
11 $\frac{R}{S} = \frac{R}{\exists} \left(\frac{S}{\exists} \right)^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$, $E \triangleq \frac{\exists \cdot}{\exists}$	$E(R) : E A \rightarrow E B$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{\exists R}{\exists} = \frac{S R}{\exists}$	absorption
15 subset $\triangleq \exists/\exists : E A \rightarrow E A$	$xs \text{ subset } ys \iff \forall a. ys \exists a \rightarrow xs \exists a$, that is $ys \subseteq xs$, not $xs \subseteq ys$.

§9.1. $i \dashv E$ Power Allegory defined as adjunction

(9.1a)



$$\frac{R}{\exists} \exists = R$$

$E A$ is the powerset of A . B&M write $P A$ — standard mathematics, but here P is already the relator $P(R)$.

§10. Relator

(10a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \subseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(R \circ S) = F(R) \circ F(S) \quad R \subseteq S \implies F(R) \subseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

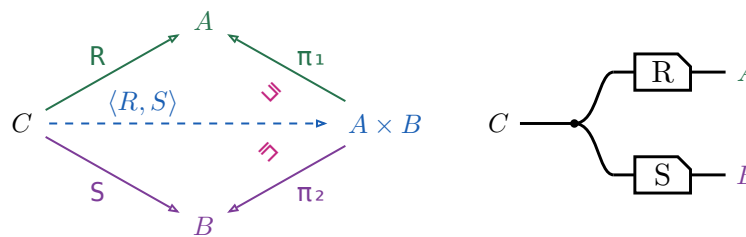
the statement	B&dM	(10b)
For f a map, $F(f)$ is a map and $F(f^\circ) = F(f)^\circ$.	Lemma 5.1	
Over a tabular allegory a functor is a relator $\iff$ it preserves $\circ$.	Theorem 5.1	
$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.	after Theorem 5.1, p. 113	
Two relators agreeing on maps are equal.	Corollary 5.1	
$F(X \cap Y) = F(X) \cap F(Y)$ for X, Y coreflexive.	Ex 5.2	
$F(R \cap S) \subseteq F(R) \cap F(S)$, and strictly.	Ex 5.2, the restriction	
$F(\text{Dom } R) = \text{Dom}(F(R))$ for F preserving $\circ$.	—	

The **power relator** $P \multimap xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§11. Fork $\langle R, S \rangle$

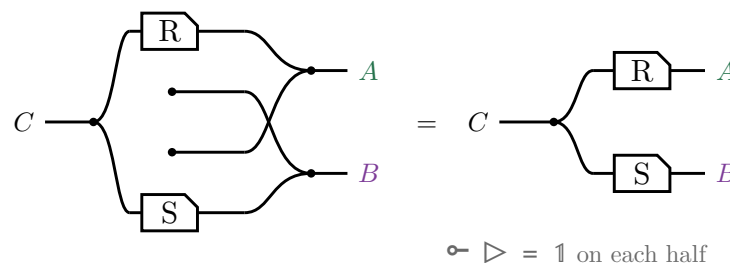
The **fork** of $R : C \rightarrow A$ and $S : C \rightarrow B$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

$$\langle R, S \rangle \pi_1 = (\text{Dom } S) R \quad \langle R, S \rangle \pi_2 = (\text{Dom } R) S$$



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \subseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In **Rel**, $c \ \langle R, S \rangle \ (a, b)$ iff $c \ R \ a$ and $c \ S \ b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

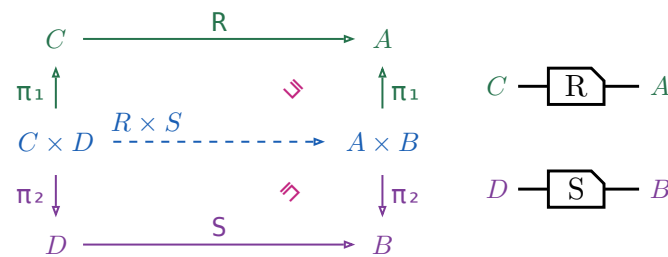
No $\circ$ survives the translation. $\pi_1 = 1 \otimes \multimap$ discards the second component, so $\pi_1^\circ = 1 \otimes \multimap$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



§11.1. Relational product $R \times S$

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

(11.1b)

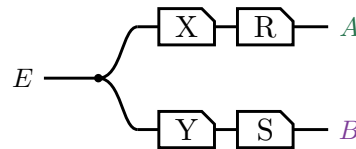


Right-then-up is $(R \times S) \pi_1$, up-then-right is $\pi_1 R$, and $(R \times S) \pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§11.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle X R, Y S \rangle$. Both sides are this picture:

(11.2a)



§12. Coproduct $[R, S] : A + B \rightarrow C$

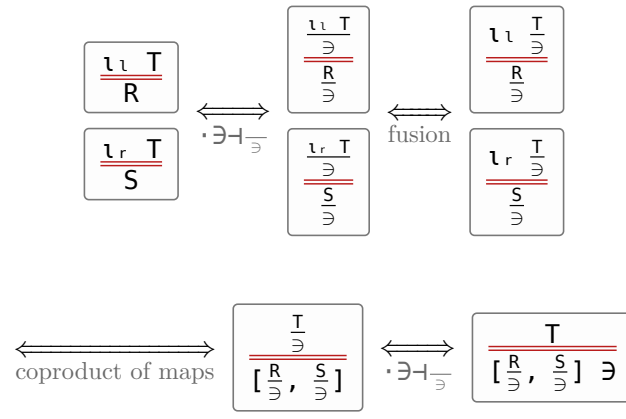
(12a)

the statement	picture
$[R, S] \triangleq \iota_l^\circ R \cup \iota_r^\circ S$ The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\frac{R}{\boxplus}, \frac{S}{\boxplus}] \ni$	
$R + S \triangleq [R \iota_l, S \iota_r]$	
$\iota_l [R, S] = R$, $\iota_r [R, S] = S$, and $[R, S]$ is the only such arrow	
$\iota_l \iota_l^\circ = \mathbb{1} = \iota_r \iota_r^\circ$	
$\iota_l \iota_r^\circ = \perp = \iota_r \iota_l^\circ$	
$\iota_l^\circ \iota_l \cup \iota_r^\circ \iota_r = \mathbb{1}$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$	

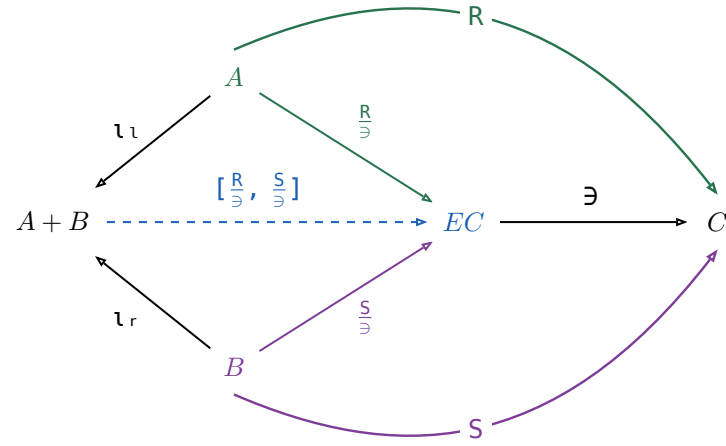
§12.1. $[R, S] \triangleq [\frac{R}{\boxplus}, \frac{S}{\boxplus}] \ni$

The universal-property row is not free: ι_l , ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\frac{\boxplus}{\boxplus}$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow E \rightarrow C$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \rightarrow C$,

(12.1a)



(12.1b)



Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = \left[\frac{R}{\exists}, \frac{S}{\exists} \right] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\iota_l \circ \frac{R}{\exists} \cup \iota_r \circ \frac{S}{\exists}) \exists = \iota_l \circ R \cup \iota_r \circ S$.

§13. The power relator $P(R)$

For $R : A \rightarrow B$,

$$P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : E A \rightarrow E B$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R)/\exists) \cap (\exists/(\exists R))^\circ$$

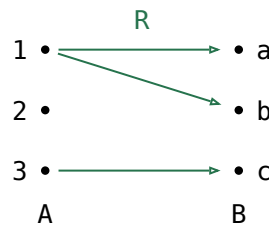
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of xs is related by R to some element of ys , and conversely.

(13a)

	in Rel	in words	$im(xs)$	partner
$(\exists R)/\exists$	$\forall y \in ys. \exists x \in xs. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$	$ys \subseteq im(xs)$	every y has a partner
$(\exists/(\exists R))^\circ$	$\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$	every y with some $xs \ R \ y$ is in ys	$ys \supseteq im(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x \ R \ y\}$	$ys = \text{every } y \text{ with some } xs \ R \ y$	$ys = im(xs)$	every y has a partner, and none is left out
$((\exists R^\circ)/\exists)^\circ$	$\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall x. x \ R \ \text{some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$	$\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$	—	every x and every y has a partner

(13b)



(13c)

xs	$(\exists R)/\exists$	$(\exists/(\exists R))^\circ$	$E(R)$	$((\exists R^\circ)/\exists)^\circ$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

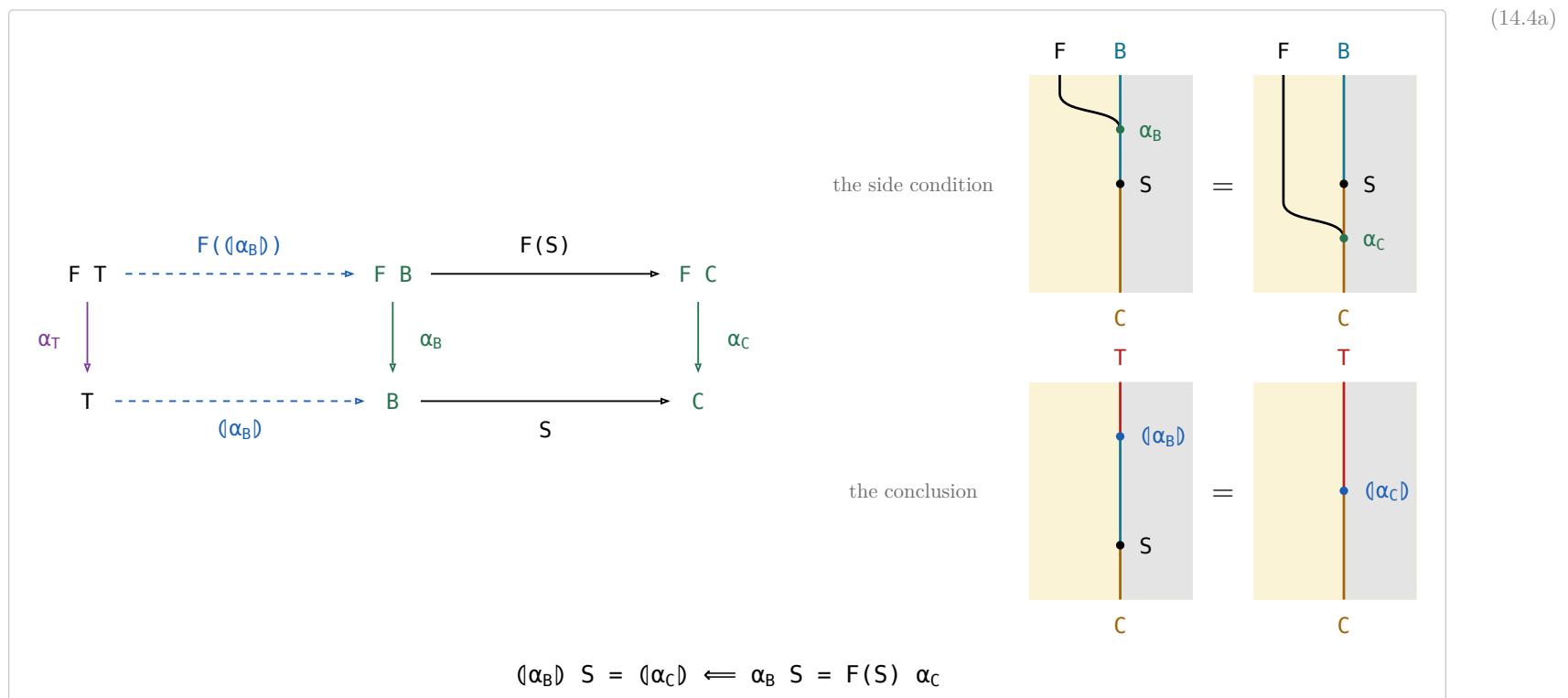
law	the reading
$X \sqsubseteq P(R) \iff X \exists \sqsubseteq \exists R \text{ and } X^\circ \exists \sqsubseteq \exists R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(1) = \frac{\exists}{\exists} = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$P(f) = \frac{\exists f}{\exists}$, for f a map	In Rel, $xs \ P(f) \ ys \iff ys = \{f \ a \mid a \in xs\}$. The half at f° says every $a \in xs$ has its $f \ a$ on ys ; f has just the one image per a , so that already says ys contains everything xs reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R \ S) = P(R) \ P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R \ S)$.

(13d)

Taking a value apart with α_T and putting it straight back is doing nothing.

§14.4. Fusion

Fusion rewrites $(\alpha_B) S$ through a second algebra $\alpha_C : F C \rightarrow C$ along an arrow $S : B \rightarrow C$.



The left square is (α_B) 's own defining square and the right one is the side condition, so the outer rectangle says $(\alpha_B) S$ satisfies the defining equation of (α_C) — and uniqueness finishes it. A fold followed by S has collapsed into a single fold, which is how an intermediate structure is got rid of.

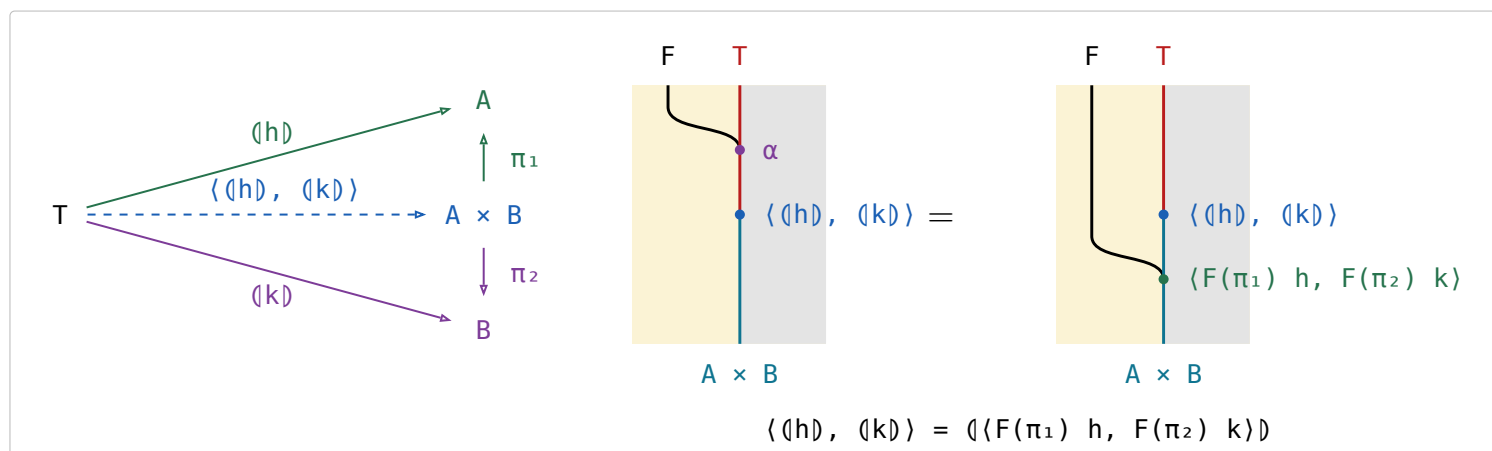
The side condition is (14.1a)'s picture again under $\alpha_T \mapsto \alpha_B$, $(\alpha_B) \mapsto S$, $\alpha_B \mapsto \alpha_C$: both say that the bead between the two merges is a homomorphism of F -algebras.

§14.5. Reduce in $\mathcal{Set}$

(14.5a)

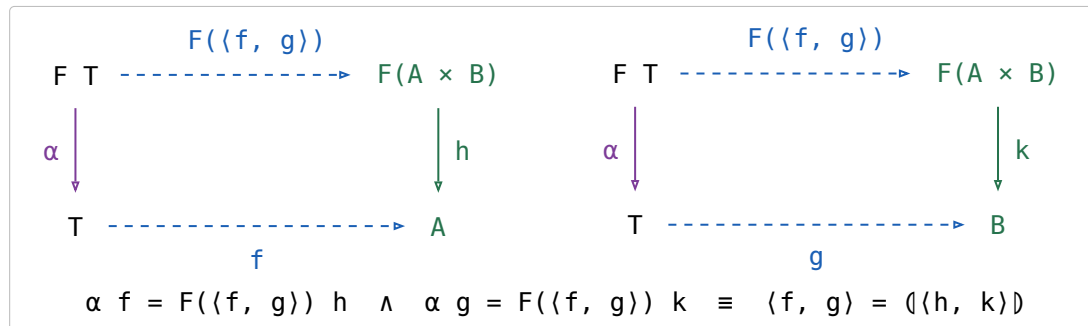
definition	type	note
$\text{listr } A ::= \text{nil} \mid \text{cons } (A, \text{listr } A)$	$\text{listr} : \text{Set} \rightarrow \text{Set}$	The cons-lists over A , the datatype every row below folds (B&dM p. 55).
$\text{sum} \triangleq \llbracket [\text{zero}, \text{plus}] \rrbracket$	$\text{sum} : \text{listr Nat} \rightarrow \text{Nat}$	$\text{plus}(a, b) = a + b$.
$\text{length} \triangleq \llbracket [\text{zero}, \pi_2 \text{ succ}] \rrbracket$	$\text{length} : \text{listr } A \rightarrow \text{Nat}$	π_2 drops the head and keeps the count of the tail, succ adds one for the head.
$\text{average} \triangleq (\text{sum}, \text{length}) \text{ div}$	$\text{average} : \text{listr Nat} \rightarrow \text{Real}$	$\text{div}(m, n) = m/n$, with $\text{div}(0, 0) = 0$ so average is total. Traverses the list twice.
banana-split law $\langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle = \llbracket (F(\pi_1) h, F(\pi_2) k) \rrbracket$	$\langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle : T \rightarrow A \times B$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to $\alpha \langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle = F(\langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle) \langle F(\pi_1) h, F(\pi_2) k \rangle$	$\alpha \langle \llbracket h \rrbracket, \llbracket k \rrbracket \rangle : F T \rightarrow A \times B$	All that (14.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.
the instance $\langle \text{sum}, \text{length} \rangle = \llbracket [\text{zeros}, \text{pluss}] \rrbracket$	$\langle \text{sum}, \text{length} \rangle : \text{listr Nat} \rightarrow \text{Nat} \times \text{Nat}$	$\text{zeros} = (\text{zero}, \text{zero})$ and $\text{pluss}(a, (b, n)) = (a + b, n + 1)$, so average runs in one pass.
$\text{preds } n = [n, n-1, \dots, 1]$	$\text{preds} : \text{Nat} \rightarrow [\text{Nat}]$	B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write preds as $\llbracket k \rrbracket \pi_1$.

(14.5b)



§14.5.1. Fokkinga's mutual recursion theorem

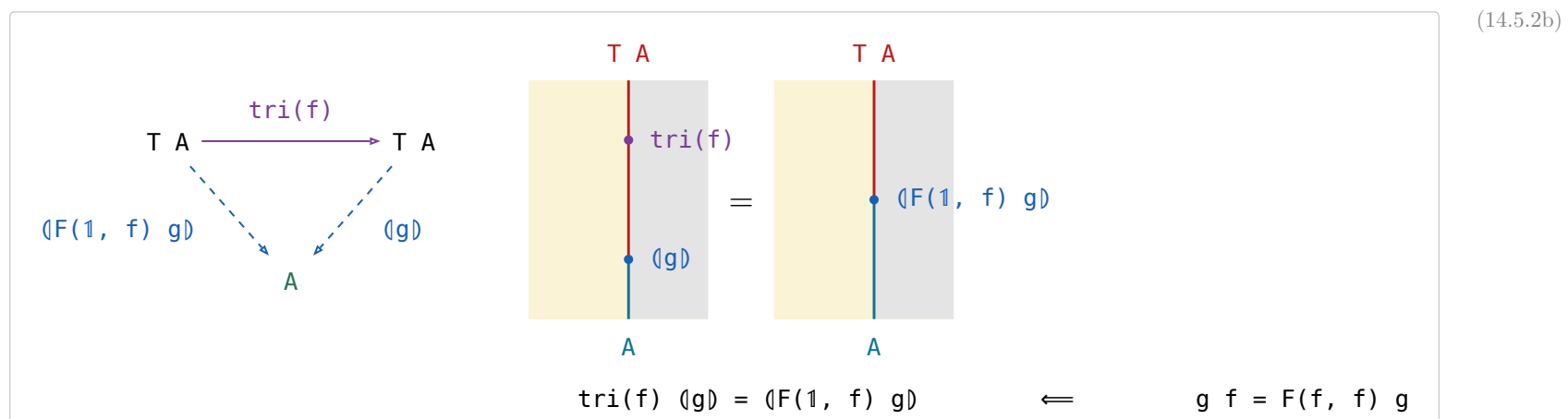
- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ creates them: the product of $h : F A \rightarrow A$ and $k : F B \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\pi_1) h, F(\pi_2) k \rangle : F(A \times B) \rightarrow A \times B$.
- π_1 and π_2 are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (14.5a) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle F(\pi_1) h, F(\pi_2) k \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in (14.5.1a) the two arrows leave $F(A \times B)$, not $F A$ and $F B$, so each sees BOTH components. The product is the case where they factor as $F(\pi_1) h$ and $F(\pi_2) k$; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



§14.5.2. Ruby triangles

stage	definition	(14.5.2a)
informally (p. 58)	$\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$	
for cons-lists (p. 59)	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{cons}] \rangle$	
with the base functor named	$F(A, B) = 1 + A \times B, \alpha = [\text{nil}, \text{cons}], \text{ so } \text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$	
abstractly (p. 59)	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$	

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : T A \rightarrow T A$.



(14.5.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$ (p. 59).

§14.5.3. Depth of a tree

the datatype	$\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$, base functor $F(A, B) = A + B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$	(14.5.3a)
the fold	$\langle [g, h] \rangle$ is the unique f with $f(\text{tip } a) = g\ a$ and $f(\text{bin } (x, y)) = h(f\ x, f\ y)$	
$\text{tree}(f)$	$\text{tree}(f) = \langle F(f, 1) \rangle [\text{tip}, \text{bin}]$; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f\ a)$ and $\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f)\ x, \text{tree}(f)\ y)$	
max	$\text{max} = \langle [1, \text{bmax}] \rangle$, where $\text{bmax } (a, b)$ is the larger of a and b	
depths	$\text{depths} = \text{tree}(\text{zero})\ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor	
depth	$\text{depth} = \text{depths}\ \text{max}$	

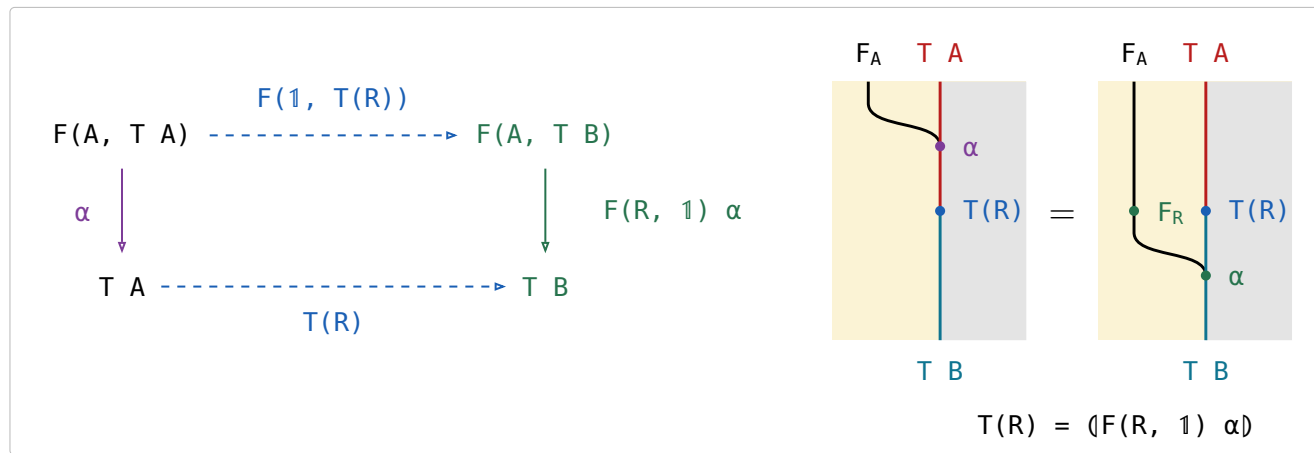
§15. Type functor

F a **binary** relator: $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, T A) \rightarrow T A$, among the maps. The **type functor** T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha) : T A \rightarrow T B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, $\text{list}(R) = ([\text{nil}, (R \otimes 1) \text{ cons}])$, which is $\text{map}(R)$.

A bifunctor puts its two arguments in two different places. The wire is the partial application $F_A = F(A, -)$, and an arrow in the SECOND argument is a bead on the object wire with the F wire running past it — the reduce section's rule, unchanged — while an arrow $R : A \rightarrow B$ in the FIRST induces a natural transformation $F_R : F_A \Rightarrow F_B$, which is a bead ON the F wire. Marsden draws bifunctors exactly so ([CatString.pdf](#), **Bifunctors**): side-by-side is already spent on functor composition, so one argument has to go into the wire's name.



$T(R)$ is the unique arrow making it commute; there is no $\sqsubseteq$ in it.

Two beads at one height on two different wires are one action, here $F(R, T(R))$, so the right panel reads $F(R, T(R)) \alpha$ and the left one $\alpha T(R)$. Their relative height says nothing: slide F_R below the $T(R)$ bead and the panel reads $F(1, T(R)) F(R, 1)$ and then α , which is the square's top row followed by its right vertical; slide it above and the panel reads $F(R, 1) F(1, T(R))$, the route through $F(B, T A)$ the square leaves out. Identifying those two is the exchange condition every bifunctor satisfies (Hinze & Marsden, Ex 1.23), and in this calculus there is nothing to identify.

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way; that is $\text{map}(R)$.
functor	$T(1) = 1$ and $T(R) T(S) = T(R S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$T(R) (Q) = (F(R, 1) Q)$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha T(R) = F(R, T(R)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion (14.4a) applied to $T(R)$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R, 1) F(1, (Q)) = F(R, (Q)) = F(1, (Q)) F(R, 1)$ — so it too needs only the initial algebra and the defining equation.

$$\begin{array}{ccc}
 F(A, T A) & \xrightarrow{F(R, T(R))} & F(B, T B) \\
 \downarrow \alpha & & \downarrow \alpha \\
 T A & \xrightarrow{T(R)} & T B
 \end{array}$$

It commutes strictly, and it is (15b) with the right vertical's two halves separated: push $F(R, 1)$ up into the top row, where it meets $F(1, T(R))$ as the one action $F(R, T(R))$, and α is left as the vertical. In the string diagram that is the F_R bead sliding up past $T(R)$, and what stays put is α , one bead falling past $T(R)$ from the row above to the row below — which is what “ α is natural” means.

§16. Combinatorial functions

definition	type	note	(16a)
$[A] ::= \text{nil} \mid \text{cons } (A, [A])$	$\text{list} : \mathcal{A} \rightarrow \mathcal{A}$	B&dM's <code>list</code> under the short name it keeps from p. 125 on.	
$\text{list}(R) \triangleq ([\text{nil}, (R \otimes 1) \text{cons}])$	$\text{list}(R) : [A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.	
$\text{subseq} \triangleq ([\text{nil}, \text{cons} \cup \pi_2])$	$\text{subseq} : [A] \rightarrow [A]$	xs subseq ys : ys is xs with elements dropped — <code>cons</code> keeps the head, π_2 drops it.	
$\text{prefix} \triangleq ([\text{nil}, \text{nil} \cup \text{cons}])$ $= \text{cat}^\circ \pi_1 = \text{init}^*$	$\text{prefix} : [A] \rightarrow [A]$	ys is an initial segment of xs ; the first <code>nil</code> is where it stops early. $\text{init} \triangleq \text{snoc}^\circ \pi_1$.	
$\text{suffix} \triangleq \text{cat}^\circ \pi_2 = \text{tail}^*$	$\text{suffix} : [A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$; as a reduce it needs <code>snoc</code> -lists.	
$\text{segment} \triangleq \text{suffix prefix}$	$\text{segment} : [A] \rightarrow [A]$	A contiguous stretch of xs : a suffix, then a prefix of that.	
$\text{partition} \triangleq \text{concat}^\circ$	$\text{partition} : [A] \rightarrow [[A]]$	<code>cat</code> restricted to a non-empty first argument, so ys is a list of non-empty segments of xs .	
$\text{concat} \triangleq ([\text{nil}, \text{cat}])$	$\text{concat} : [[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.	
inits	$\text{inits} : [A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\ni}$, listing the prefixes by increasing length.	
tails	$\text{tails} : [A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order, which is what $\text{splits} \triangleq (\text{inits}, \text{tails}) \text{zip}$ needs to implement $\frac{\text{cat}^\circ}{\ni}$.	
$\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\ni} \max R$	$\text{filter}(p) : [A] \rightarrow [A]$	The longest subsequence of xs whose every element passes p . p is a coreflexive and $\text{list}(p)$ the relator applied to it, so $\text{list}(p)$ is the test “every element passes” — no boolean anywhere. The longest is unique: only one subsequence keeps exactly the passing elements. The program is $([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$. Ex 7.41; $\max R$ is (17.1a)	
$R \triangleq \text{length} \leq \text{length}^\circ$	$R : [A] \rightarrow [A]$	The preorder <code>filter</code> and <code>takewhile</code> maximise over: the longer list wins.	
$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\ni} \max R$	$\text{takewhile}(p) : [A] \rightarrow [A]$	The same with <code>prefix</code> for <code>subseq</code> : the longest prefix whose every element passes p . Unique, since the prefixes of one list are linearly ordered by length. Ex 7.39	
$\text{mss} \triangleq \frac{\text{segment sum}}{\ni} \max \leq$	$\text{mss} : [\text{Int}] \rightarrow \text{Int}$	Maximum segment sum. <code>segment = suffix prefix</code> splits it into $\frac{\text{prefix sum}}{\ni} \max \leq$ on each suffix, which the greedy theorem turns into a reduce. Ex 7.40	

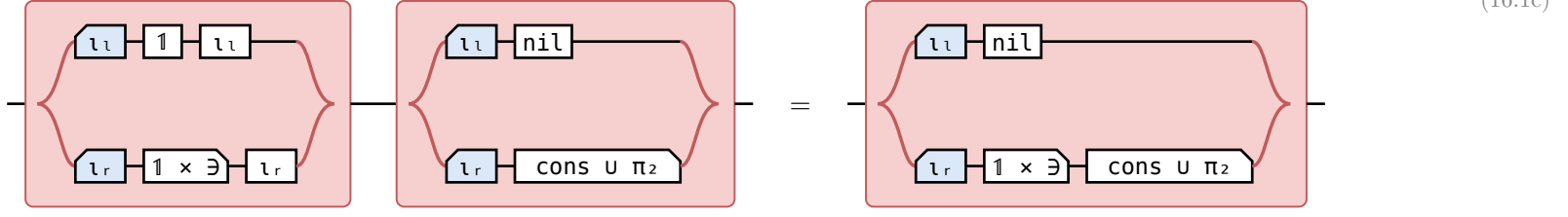
§16.1. $\frac{\text{subseq}}{\ni} = ([\text{nil}, \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} E(\text{cons}), \pi_2) \text{cup}])$

$$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : E A \times E A \rightarrow E A, \text{ so } \frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{cup}.$$

(16.1a)

$$\frac{F(\exists) \frac{[nil, cons \cup \pi_2]}{\exists}}{F X = 1 + A \times X} \xlongequal{\text{coproduct}} \frac{(1 + 1 \times \exists) \frac{[nil, cons \cup \pi_2]}{\exists}}{\exists} \xlongequal{\text{coproduct}} \frac{[nil, (1 \times \exists)(cons \cup \pi_2)]}{\exists} \quad (16.1b)$$

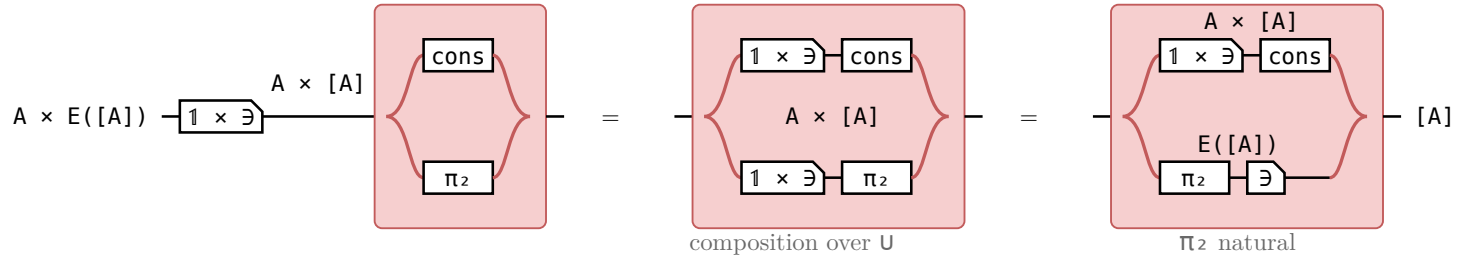
$$\xlongequal{\text{coproduct of maps}} \left[\frac{nil}{\exists}, \frac{(1 \times \exists)(cons \cup \pi_2)}{\exists} \right] \xlongequal{\text{singleton}} \left[nil \frac{1}{\exists}, \frac{(1 \times \exists)(cons \cup \pi_2)}{\exists} \right]$$



B&dM p. 124, “coproduct”: $R + S \triangleq [R \text{ } l_l, S \text{ } l_r]$ and $l_l [R, S] = R, l_r [R, S] = S$ — (12a).

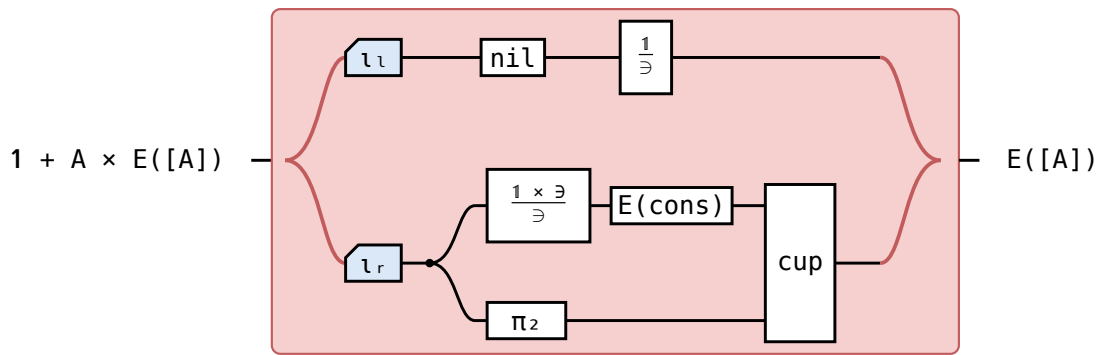
$$\frac{(1 \times \exists)(cons \cup \pi_2)}{\exists} \xlongequal{\text{U composes, } \pi_2 \text{ natural}} \frac{(1 \times \exists) cons \cup \pi_2 \exists}{\exists} \xlongequal{\text{cup}} \left\langle \frac{(1 \times \exists) cons}{\exists}, \frac{\pi_2 \exists}{\exists} \right\rangle \text{ cup} \quad (16.1d)$$

$$\xlongequal{\text{absorption, fusion}} \left\langle \frac{1 \times \exists}{\exists} E(cons), \pi_2 \right\rangle \text{ cup}$$



B&dM p. 124, “composition over join, naturality of π_2 ”: $T(X_1 \cup X_2) = T X_1 \cup T X_2$ — (1.2b); $(1 \times \exists) \pi_2 = \pi_2 \exists$ — (11.1b) at π_2 , an equality because 1 is entire.

$$\begin{array}{ccc} A \times E([A]) & \xrightarrow{\pi_2} & E([A]) \\ \downarrow 1 \times \exists & & \downarrow \exists \\ A \times [A] & \xrightarrow{\pi_2} & [A] \end{array}$$



B&dM §5.6, p. 124, which writes $Pcons$; $cons$ is a map, and there $P(cons) = E(cons)$ — (13d).

§16.2. $takewhile(p) = ([nil, (\pi_1 p \rightarrow cons, \rightarrow nil)])$

$F X = 1 + A \times X$, $\alpha \triangleq [nil, cons]$, p a coreflexive on A , $R \triangleq length \leq length^\circ$, a preorder, $S \triangleq [nil, \rightarrow nil \cup (p \times 1) cons] : F([A]) \rightarrow [A]$.

$\rightarrow$ nil is the constant nil, the second nil of $\text{prefix} = ([\text{nil}, \rightarrow \text{nil} \cup \text{cons}])$; $(g \rightarrow X, Y)$ is X where g is defined and Y where it is not.

$\text{nil } R = \top$ nil is the shortest list, so it loses every $\max R$

$$\boxed{\alpha \text{ prefix list}(p)} \xrightarrow{\text{defining equation}} \boxed{F(\text{prefix}) [\text{nil}, \rightarrow \text{nil} \cup \text{cons}] \text{ list}(p)} \quad (16.2b)$$

$$\boxed{\text{list}(p) \text{ through cons}} \xrightarrow{\text{defining equation}} \boxed{F(\text{prefix}) [\text{nil}, \rightarrow \text{nil} \cup (p \times \text{list}(p)) \text{ cons}]}$$

$$\boxed{\text{relator, prefix entire}} \xrightarrow{\text{defining equation}} \boxed{[\text{nil}, \rightarrow \text{nil} \cup (p \times (\text{prefix list}(p))) \text{ cons}]} \xrightarrow{\text{prefix list}(p) \text{ entire}} \boxed{F(\text{prefix list}(p)) S}$$

(14.1a) reads that off as $\text{prefix list}(p) = ([S])$. (14.4a) cannot: $\text{list}(p)$ is not entire, $(1 \times \text{list}(p)) \rightarrow \text{nil} \sqsubset \rightarrow \text{nil}$, and no algebra meets the side condition.

$$\boxed{(1 \times R^\circ)(\rightarrow \text{nil} \cup (p \times 1) \text{ cons})} \xrightarrow{\text{relator, composition over } U} \boxed{(1 \times R^\circ) \rightarrow \text{nil} \cup (p \times R^\circ) \text{ cons}} \xrightarrow{(5.1a), (1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons } R^\circ} \boxed{\rightarrow \text{nil} \cup (p \times 1) \text{ cons } R^\circ} \xrightarrow{R^\circ \text{ reflexive}} \boxed{(\rightarrow \text{nil} \cup (p \times 1) \text{ cons}) R^\circ} \quad (16.2c)$$

the cons branch of $F(R^\circ) S \sqsubseteq S R^\circ$; the nil branch is $\text{nil} \sqsubseteq \text{nil } R^\circ$. $(1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons } R^\circ$ is $\text{cons length} = (1 \times \text{length}) \pi_2 \text{ succ}$ with succ monotone — a longer tail makes a longer list.

$$\boxed{\frac{S}{\exists} \max R} \xrightarrow{\text{coproduct of maps}} \boxed{[\frac{\text{nil}}{\exists} \max R, \frac{\rightarrow \text{nil} \cup (p \times 1) \text{ cons}}{\exists} \max R]} \xrightarrow{\text{singleton, } R^\circ \text{ reflexive}} \boxed{[\text{nil}, \frac{\rightarrow \text{nil} \cup (p \times 1) \text{ cons}}{\exists} \max R]} \quad (16.2d)$$

$$\boxed{\text{nil } R = \top} \xrightarrow{\text{defining equation}} \boxed{[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})]}$$

the set is $\{\text{nil}\}$ where p fails on the head and $\{\text{nil}, \text{cons}(a, xs)\}$ where it holds, and nil loses the second — (17.1a) at a two-element set.

the law	what it says	(16.2e)
$(\frac{S}{\exists} \max R) \sqsubseteq \frac{([S])}{\exists} \max R$ Theorem 7.2, (17.3e) at R° ; $\max R = \min R^\circ$, so the condition is (16.2c)	greedy: one longest p -prefix kept at each cons , instead of every p -prefix collected and one chosen at the end	
$\text{prefix}^\circ \text{ prefix } n R n R^\circ \sqsubseteq 1$ context, (17.1b): only R between prefixes of one list counts	two prefixes of one list of equal length are equal, so $\frac{\text{prefix list}(p)}{\exists} \max R$ is simple — the longest, not a longest	
$\frac{\text{prefix list}(p)}{\exists} \max R$ entire	nil is always a p -prefix and R is connected on the prefixes of one list, so the longest exists	
$X \sqsubseteq Y, X \text{ entire}, Y \text{ simple} \implies X = Y$ $([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$ is a reduce of maps, hence entire	what turns the greedy $\sqsubseteq$ into the heading's =	

§16.3. $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$

F, α, p, R as in (16.2a); $S \triangleq [\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}] : F([A]) \rightarrow [A], \pi_2$ where (16.2a) has $\rightarrow \text{nil}$.

$1 \sqsubseteq \pi_2 R \text{ cons}^\circ$ the tail is one shorter than the cons , so π_2 loses the $\max R$ at every step — where (16.2a)'s loser is nil

$$\boxed{\alpha \text{ subseq list}(p)} \xrightarrow{\text{defining equation}} \boxed{F(\text{subseq}) [\text{nil}, \text{cons } \cup \pi_2] \text{ list}(p)} \xrightarrow{\text{list}(p) \text{ through cons, } \pi_2 \text{ natural}} \boxed{F(\text{subseq}) [\text{nil}, (p \times \text{list}(p)) \text{ cons } \cup (1 \times \text{list}(p)) \pi_2]} \quad (16.3b)$$

$$\boxed{\text{relator, subseq list}(p) \text{ entire}} \xrightarrow{\text{defining equation}} \boxed{F(\text{subseq list}(p)) S}$$

(14.1a) reads that off as $\text{subseq list}(p) = \langle S \rangle$ — (16.2b)’s chain with π_2 for $\rightarrow \text{nil}$. Here (14.4a) is blocked for the same reason, and π_2 is the branch that carries $\text{list}(p)$ out by naturality ((16.1f) at $\exists := \text{list}(p)$).

$$\boxed{(\mathbb{1} \times R^\circ)(\pi_2 \cup (\text{p} \times \mathbb{1}) \text{ cons})} \xrightarrow[\pi_2 \text{ natural}]{\text{composition over } \cup,} \boxed{\pi_2 R^\circ \cup (\text{p} \times R^\circ) \text{ cons}} \xrightarrow[(16.2c)\text{'s cons branch}]{\subseteq} \boxed{\pi_2 R^\circ \cup (\text{p} \times \mathbb{1}) \text{ cons } R^\circ} \xrightarrow[\text{composition over } \cup]{=} \boxed{(\pi_2 \cup (\text{p} \times \mathbb{1}) \text{ cons}) R^\circ} \quad (16.3c)$$

$F(R^\circ) S \sqsubseteq S R^\circ$, the nil branch again $\text{nil} \sqsubseteq \text{nil } R^\circ$. The first step is an equality: (16.2c) spends R° reflexive on getting $\rightarrow \text{nil}$ past $\mathbb{1} \times R^\circ$, and π_2 needs only its naturality square.

$$\boxed{\frac{S}{\exists} \max R} \xrightarrow[(16.2d)\text{'s first two steps}]{=} \boxed{[\text{nil}, \frac{\pi_2 \cup (\text{p} \times \mathbb{1}) \text{ cons}}{\exists} \max R]} \xrightarrow[\mathbb{1} \sqsubseteq \pi_2 R \text{ cons}^\circ]{=} \boxed{[\text{nil}, (\pi_1 \text{ p} \rightarrow \text{cons}, \pi_2)]} \quad (16.3d)$$

at (a, xs) the set is $\{xs\}$ where p fails on a and $\{xs, \text{cons}(a, xs)\}$ where it holds, and xs loses the second — (17.1a) at a two-element set. The head is dropped, not the whole tail: that is the one place π_2 shows against (16.2d)’s $\rightarrow \text{nil}$.

the law	what it says
$\langle \frac{S}{\exists} \max R \rangle \sqsubseteq \langle \frac{\langle S \rangle}{\exists} \max R \rangle$ (16.2e)’s first row at this S , its condition (16.3c)	greedy: one longest p -subsequence kept at each cons
$S^\circ S \cap R \cap R^\circ \sqsubseteq \mathbb{1}$ fails at $S := \text{subseq list}(p)$ (16.2e)’s uniqueness row does not transfer	two p -subsequences of one list can have equal length and differ: the prefixes of one list are linearly ordered by length, its subsequences are not
$\frac{\text{subseq list}(p)}{\exists} \max R$ simple (16a)	a p -subsequence that drops a passing element is beaten by the one that keeps it, so only the subsequence keeping exactly the passing elements survives $\max R$ — the longest, not a longest
$X \sqsubseteq Y$, X entire, Y simple $\implies X = Y$ (16.2e)’s last row at this S	what turns the greedy $\sqsubseteq$ into the heading’s $=$; $\langle [\text{nil}, (\pi_1 \text{ p} \rightarrow \text{cons}, \pi_2)] \rangle$ is again a reduce of maps, hence entire

§16.4. $\text{mss} = \langle [\text{zero wrap}, ((\mathbb{1} \times \text{head}) \oplus, \pi_2) \text{ cons}] \rangle \max \leq$

$F X = \mathbb{1} + A \times X$, $A := \text{Int}$, $\alpha \triangleq [\text{nil}, \text{cons}]$, $\text{sum} = \langle [\text{zero}, \text{plus}] \rangle$ and $\text{segment} = \text{suffix prefix}$ from (14.5a) and (16a).

$\text{head} \triangleq \text{cons}^\circ \pi_1$, $\text{wrap} \triangleq (\mathbb{1}, \rightarrow \text{nil}) \text{ cons}$ the head of a list and the one-element list, beside (16a)’s $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\rightarrow \text{zero} \cup \text{plus}}{\exists} \max \leq$ B&dM’s $\text{oplus} = \max \wedge (\text{zero} \cup \text{plus})$; the set at (a, b) is $\{0, a + b\}$, so $\oplus$ is the larger of the two

$$\boxed{\frac{\text{segment sum}}{\exists} \max \leq} \xrightarrow[\text{segment} = \text{suffix prefix}]{=} \boxed{\frac{\text{suffix (prefix sum)}}{\exists} \max \leq} \xrightarrow[\text{absorption}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\text{prefix sum}) \max \leq} \quad (16.4a)$$

$$\boxed{\frac{R}{\exists} \exists = R, \text{union} = E(\exists)} \xrightarrow[\text{suffix } E(\frac{\text{prefix sum}}{\exists}) \text{ union } \max \leq]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists}) \text{ union } \max \leq} \xrightarrow[(17.1b), \text{the sets non-empty}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists}) E(\max \leq) \max \leq}$$

$$\xrightarrow[\text{relator}]{=} \boxed{\frac{\text{suffix}}{\exists} E(\frac{\text{prefix sum}}{\exists} \max \leq) \max \leq}$$

B&dM’s $\max P(\max \wedge (\text{sum prefix})) \wedge \text{suffix}$, mirrored. The union step is (17.1b)’s $P(\min R) \min R = P(\text{Dom}(\min R)) \text{ union } \min R$ — every suffix has the empty prefix, so Dom is $\mathbb{1}$ here — and $P(f) = E(f)$ at the map it is applied to ((13d)).

$$\boxed{[\text{nil}, \rightarrow \text{nil} \cup \text{cons}] \text{ sum}} \xrightarrow[\text{coproduct of maps, composition over } \cup]{=} \boxed{[\text{nil sum}, \rightarrow \text{nil sum} \cup \text{cons sum}]} \xrightarrow[\text{sum's defining equation}]{=} \boxed{[\text{zero}, \rightarrow \text{zero} \cup (\mathbb{1} \times \text{sum}) \text{ plus}]} \quad (16.4c)$$

$$\xrightarrow[\text{relator}]{=} \boxed{(\mathbb{1} \times \text{sum}) \rightarrow = \rightarrow, \text{sum entire}} \boxed{[\text{zero}, (\mathbb{1} \times \text{sum})(\rightarrow \text{zero} \cup \text{plus})]} \xrightarrow[\text{relator}]{=} \boxed{F(\text{sum}) [\text{zero}, \rightarrow \text{zero} \cup \text{plus}]}$$

(14.4a) at $\alpha_B := [\text{nil}, \multimap \text{nil} \cup \text{cons}], S := \text{sum}$: the side condition, so $\text{prefix sum} = ([\text{zero}, \multimap \text{zero} \cup \text{plus}])$. prefix is the reduce, sum the map fused into it — the intermediate list is gone.

$$\begin{array}{c}
 \boxed{(1 \times \leq^\circ)(\multimap \text{zero} \cup \text{plus})} \xrightarrow[\text{over } U]{\text{relator, composition}} \boxed{(1 \times \leq^\circ) \multimap \text{zero} \cup (1 \times \leq^\circ) \text{plus}} \quad (5.1a), (\leq^\circ \times \leq^\circ) \text{plus} \sqsubseteq \text{plus} \leq^\circ \quad \boxed{\multimap \text{zero} \cup \text{plus} \leq^\circ} \xrightarrow[\text{reflexive}]{\leq^\circ} \boxed{(\multimap \text{zero} \cup \text{plus}) \leq^\circ}
 \end{array} \tag{16.4d}$$

the plus branch of $F(\leq^\circ) S \sqsubseteq S \leq^\circ$; the zero branch is $\text{zero} \sqsubseteq \text{zero} \leq^\circ$. $(\leq \times \leq) \text{plus} \sqsubseteq \text{plus} \leq$ is (17.3a), written $+$ there, and (17.3e)'s first row carries it to $\leq^\circ$ — plus is a map, so it is monotonic on an order and on its opposite together.

$$\boxed{\frac{[\text{zero}, \multimap \text{zero} \cup \text{plus}]}{\exists} \text{max} \leq} \xrightarrow[\text{coproduct of maps}]{} \boxed{[\frac{\text{zero}}{\exists} \text{max} \leq, \frac{\multimap \text{zero} \cup \text{plus}}{\exists} \text{max} \leq]} \xrightarrow[\text{singleton, } \leq^\circ \text{ reflexive}]{} \boxed{[\text{zero}, \oplus]} \tag{16.4e}$$

with (16.4d) the greedy theorem gives $([\text{zero}, \oplus]) \sqsubseteq \frac{\text{prefix sum}}{\exists} \text{max} \leq$ — B&dM's own containment — and (16.4g)'s second row makes it an equality.

$$\boxed{[\text{nil wrap}, ((1 \times \text{head}) \text{cons}, \pi_2) \text{cons}] \text{list}(g)} \xrightarrow[\text{coproduct of maps, wrap and cons natural}]{} \boxed{[\text{nil g wrap}, ((1 \times \text{head}) \text{cons g}, \pi_2 \text{list}(g)) \text{cons}]} \tag{16.4f}$$

$$\xrightarrow[\text{head g}]{\text{g's defining equation, list}(g) \text{ head} =} \boxed{[c \text{ wrap}, ((1 \times (\text{list}(g) \text{ head})) f, (1 \times \text{list}(g)) \pi_2) \text{cons}]} \xrightarrow[\text{fork, relator}]{} \boxed{F(\text{list}(g)) [c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{cons}]}$$

$g \triangleq ([c, f])$ and $\text{tails} = ([\text{nil wrap}, ((1 \times \text{head}) \text{cons}, \pi_2) \text{cons}])$, since $\text{tails}(\text{cons}(a, \text{xs}))$ is $\text{cons}(\text{cons}(a, \text{head}(\text{tails xs})), \text{tails xs})$. (14.4a) then reads off $\text{tails list}(g) = ([c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{cons}])$, at $c, f := \text{zero}, \oplus$ the heading's fold.

the law	what it says	(16.4g)
$(\frac{S}{\exists} \text{max} \leq) \sqsubseteq \frac{(S)}{\exists} \text{max} \leq$ (17.3e) at $\leq^\circ, S := [\text{zero}, \multimap \text{zero} \cup \text{plus}]$, condition (16.4d)	greedy: one running maximum kept at each cons , instead of every prefix sum collected and one chosen at the end	
$X \sqsubseteq Y, X \text{ entire}, Y \text{ simple} \implies X = Y$ (16.2e)'s last row at this S	$([\text{zero}, \oplus])$ is a reduce of maps, and the prefix sums of one list are finitely many integers with one greatest, so (16.4e)'s $\sqsubseteq$ is an equality	
$\text{list}(g) \text{ head} = \text{head } g$ the one step of (16.4f) the note has no law for	head is a partial map, undefined at nil ; the equality holds because tails never returns an empty list, which is a fact about tails , not a naturality square	
tails implements $\frac{\text{suffix}}{\exists}$, $\text{list}(f)$ implements $E(f)$ (16a)	what makes (16.4b) a program: one fold builds the n+1 running maxima, and the final max $\leq$ reads them in one more pass, so mss is linear	

§17. Optimisation Problems

§17.1. Minimum and maximum

(17.1a)

For $R : A \rightarrow A$, $\min R \triangleq \exists n (\epsilon \backslash R^\circ) : E A \rightarrow A$, $\max R \triangleq \min R^\circ$.

$xs (\min R) x \iff x \in xs \wedge (\forall y \in xs. x R y)$

$xs (\min R) x \iff (x \text{ in } xs) \text{ and all } x R \backslash xs \text{ in } q$

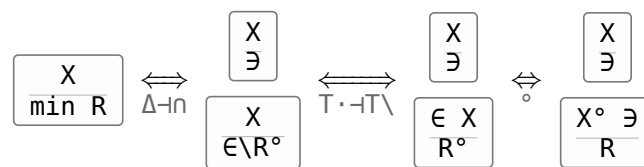
$\min R = \exists n \text{ all } R^\circ \text{ all } R \triangleq \epsilon \backslash R, q's \text{ all};$ the chains below keep it written $\epsilon \backslash$

(17.1b)

the law	what it says
$X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } X^\circ \exists \sqsubseteq R$	in the set, and below every element of it
$\frac{1}{\exists} (\epsilon \backslash R) = R$	bounding a singleton is bounding its element
$\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$	bound S 's image without building the set
$\text{union} \triangleq \frac{\exists \exists}{\exists} : E(E A) \rightarrow E A$	flattens a set of sets
$\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$	bound a union by bounding each member set
$\frac{1}{\exists} \min R = 1 n R$	a singleton's minimum is its element, where R is reflexive $\frac{S}{\exists} \min R \text{ at } S := 1$
$\frac{S}{\exists} \min R = S n (S^\circ \backslash R^\circ)$	an S -value that points to every S -value
$\frac{S}{\exists} \min R = \frac{S}{\exists} \min (R n S^\circ S)$	only R between values S gives one argument counts — context
$E(S) \min R = (\exists S) n ((\exists S)^\circ \backslash R^\circ)$	the same for the image of a set $\frac{S}{\exists} \min R \text{ at } S := \exists S$
$P(f) \min R = \min (f R f^\circ) f$	shunt a function through a minimum
$P(S) \min R = (\exists S) n (\epsilon \backslash (S R^\circ))$ $R \text{ reflexive}$	fusion with the power relator $\exists$ is the only proof here that tabulates
$P(S) \min R \sqsubseteq (\exists S) n (\epsilon \backslash (S R^\circ))$	the half of the row above that costs nothing
$P(\min R) \min R \sqsubseteq \text{union} \min R$ $R \text{ a preorder}$	a minimum in each set, then a minimum of those
$P(\min R) \min R = P(\text{Dom} (\min R)) \text{ union} \min R$ $R \text{ a preorder}$	the same as an equality, once empty sets are dropped

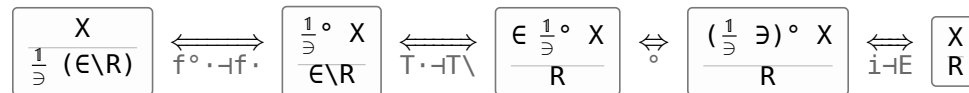
§17.1.1. $X \sqsubseteq \min R \iff X \sqsubseteq \exists \text{ and } X^\circ \exists \sqsubseteq R$

(17.1.1a)



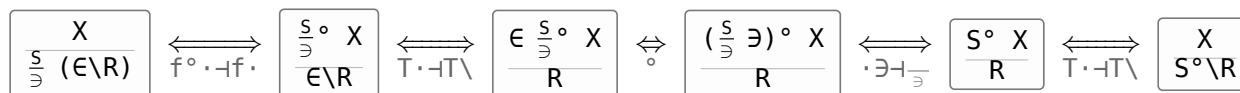
§17.1.2. $\frac{1}{\exists} (\epsilon \backslash R) = R$

(17.1.2a)



§17.1.3. $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$

(17.1.3a)



§17.1.4. $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$

(17.1.4a)



§17.1.5. $\frac{S}{\exists} \min R = S \cap (S^\circ \setminus R^\circ)$

(17.1.5a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min R}} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \boxed{\frac{\frac{S^\circ}{\exists} X}{\min R}} \xleftrightarrow{\Delta \neg \cap} \boxed{\frac{\frac{S^\circ}{\exists} X}{\exists}} \xleftrightarrow{f^\circ \cdot \neg f^\circ} \boxed{\frac{X}{\frac{S}{\exists} \exists}} \xleftrightarrow{\cdot \exists \neg \frac{S}{\exists}, (17.1.3a)} \boxed{\frac{X}{S}} \xleftrightarrow{\Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}}
 \end{array}$$

§17.1.6. $\frac{S}{\exists} \min R = \frac{S}{\exists} \min(R \cap S^\circ S)$

(17.1.6a)

$$\begin{array}{c}
 \boxed{\frac{X}{\frac{S}{\exists} \min(R \cap S^\circ S)}} \xleftrightarrow{(17.1.5a), ^\circ} \boxed{\frac{X}{S \cap (S^\circ \setminus (R^\circ \cap S^\circ S))}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{X}{S}} \xleftrightarrow{\Delta \neg \cap, S^\circ \cdot \text{monotone}} \boxed{\frac{X}{S}} \xleftrightarrow{T \cdot \neg T \setminus, \Delta \neg \cap} \boxed{\frac{X}{S \cap (S^\circ \setminus R^\circ)}} \xleftrightarrow{(17.1.5a)} \boxed{\frac{X}{\frac{S}{\exists} \min R}}
 \end{array}$$

§17.1.7. $P(f) \min R = \min(f R f^\circ) f$

(17.1.7a)

$$\begin{array}{c}
 \boxed{\frac{P(f) \min R}{f \text{ a map}}} \xleftrightarrow{P = E \text{ on maps, (17.1.5a)}} \boxed{(\exists f) \cap ((\exists f)^\circ \setminus R^\circ)} \xleftrightarrow{^\circ, T \cdot \neg T \setminus, f^\circ \cdot \neg f^\circ} \boxed{(\exists f) \cap (E \setminus (f R^\circ))} \\
 \xleftrightarrow{\text{modular law, } f \text{ simple}} \boxed{(\exists \cap ((E \setminus (f R^\circ)) f^\circ)) f} \xleftrightarrow{\cdot f \neg f^\circ, ^\circ, \min} \boxed{\min(f R f^\circ) f}
 \end{array}$$

§17.1.8. $P(S) \min R \sqsubseteq (\exists S) \cap (E \setminus (S R^\circ))$

(17.1.8a)

$$\begin{array}{c}
 \boxed{\frac{P(S) \min R}{(\exists S) \cap (E \setminus (S R^\circ))}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{P(S) \min R}{\exists S}} \xleftrightarrow{\text{UP of min}} \boxed{\frac{P(S) \exists}{\exists S}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{P(S) \exists}{\exists S}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{P(S) \exists}{\exists S}}
 \end{array}$$

§17.1.9. $P(\min R) \min R \sqsubseteq \text{union} \min R$

(17.1.9a)

$$\begin{array}{c}
 \boxed{\frac{P(\min R) \min R}{\text{union} \min R}} \xleftrightarrow{(17.1.5a)} \boxed{\frac{P(\min R) \min R}{(\exists \exists) \cap ((\exists \exists)^\circ \setminus R^\circ)}} \xleftrightarrow{^\circ, \Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{P(\min R) \min R}{\exists \exists}} \xleftrightarrow{\text{UP of min, } R \text{ transitive}} \boxed{\frac{P(\min R) \exists}{\exists \min R}} \xleftrightarrow{\Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\frac{P(\min R) \exists}{\exists \min R}}
 \end{array}$$

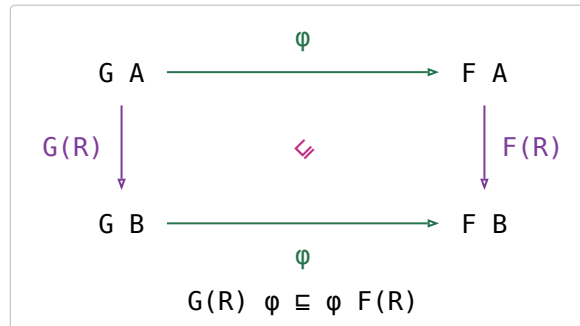
§17.2. Lax natural transformations

$\varphi_A : G A \rightarrow F A$ is a **lax natural transformation** $\varphi : G \Rightarrow F$ when, for every $R : A \rightarrow B$, $G(R) \varphi \sqsubseteq \varphi F(R)$. (5.13)

(17.2a)

$\exists : P \Rightarrow \text{Id}$ — that is $P(R) \exists \sqsubseteq \exists R$, (13d)'s first row at $X := P(R)$.

It is enough to check **maps**, and there it is an equality $G(f) \varphi = \varphi F(f)$: laxness is about relations only. Theorem 5.2



(17.2b)

§17.3. Monotonic algebras

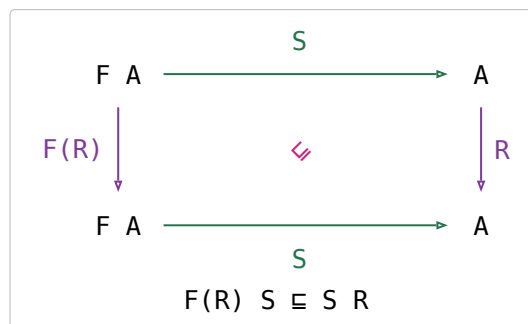
An F -algebra $S : F A \rightarrow A$ is **monotonic on** $R : A \rightarrow A$ if $F(R) S \sqsubseteq S R$.

(17.3a)

For a map $f : F A \rightarrow A$ that is $f \circ F(R) f \sqsubseteq R$, equivalently $F(R) \sqsubseteq f R f^\circ$.

$(\leq \times \leq) + \sqsubseteq + \leq$ — addition on Nat is monotonic on $\leq$, which at the point level reads $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$.

$F(R) S \sqsubseteq S R$ is (17.2a)'s inequation at $G := F$, $F := \text{Id}$, $\varphi := S$, so (17.3b) is (17.2b) with $\text{Id}(R)$ written R .

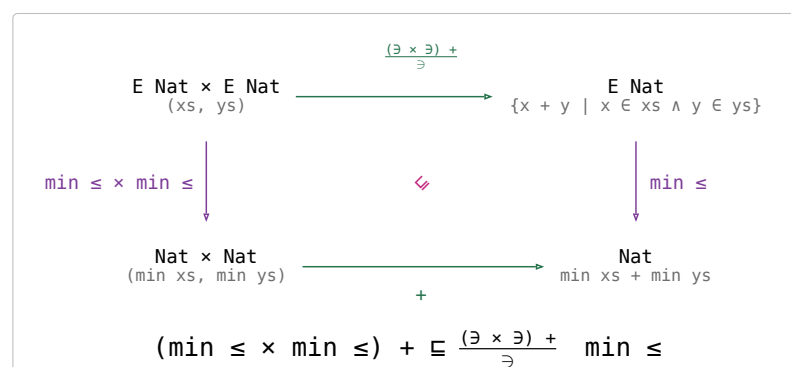
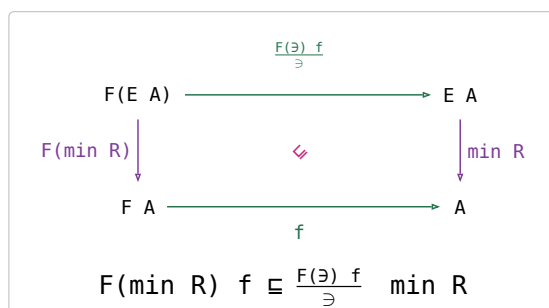


(17.3b)

$f : F A \rightarrow A$ **distributes over** R if $F(\min R) f \sqsubseteq \frac{F(\exists) f}{\exists} \min R$.

(17.3c)

$+$ distributes over $\leq$, at the point level $\min xs + \min ys = \min\{x + y \mid x \in xs \wedge y \in ys\}$ for xs, ys non-empty and $\min \triangleq \min \leq$.

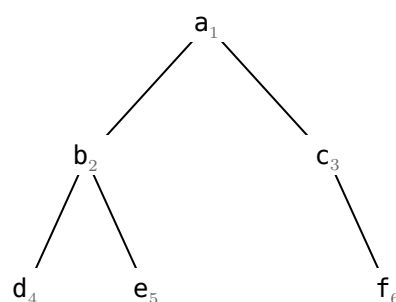
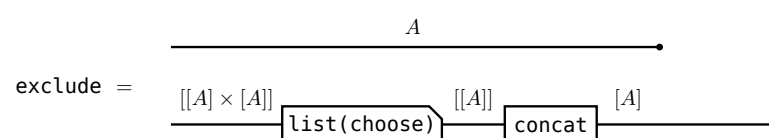
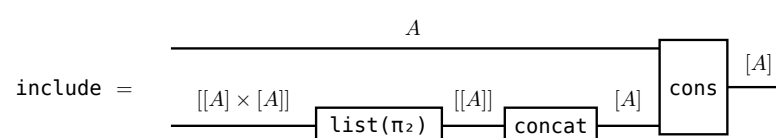


(17.3d)

the law	what it says	(17.3e)
$f^\circ F(R) f \sqsubseteq R \iff f^\circ F(R^\circ) f \sqsubseteq R^\circ$	a map is monotonic on an order and on its opposite together — a relation is not	
$f^\circ F(R) f \sqsubseteq R \iff F(\min R) f \sqsubseteq \frac{F(\exists) f}{\exists} \min R$ Theorem 7.1, f a map	for a map the two definitions above are one condition	
$(\frac{S}{\exists} \min R) \sqsubseteq \frac{(S)}{\exists} \min R$ Theorem 7.2, R a preorder and $F(R) S \sqsubseteq S R$	the greedy theorem : keeping one minimum at every step beats no more than taking the minimum at the end	

§17.4. Planning a company party

definition	type	note	(17.4a)
$\text{tree } A ::= \text{node } (A, [\text{tree } A])$	$\text{tree} : \mathcal{A} \rightarrow \mathcal{A}$	The company hierarchy: an employee, and the list of subtrees under them.	
$F(A, B) = A \times [B]$	$F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	The base functor tree folds: an employee beside the recursive position, one layer deep.	
rating	$\text{rating} : A \rightarrow \text{Real}$	What one employee is worth as a guest.	
$\text{cost} \triangleq \text{list}(\text{rating}) \text{ sum}$	$\text{cost} : [A] \rightarrow \text{Real}$	What a guest list is worth.	
$R \triangleq \text{cost} \leq \text{cost}^\circ$	$R : [A] \rightarrow [A]$	The preorder the guest list is maximised over.	
$\text{choose} \triangleq \pi_1 \cup \pi_2$	$\text{choose} : [A] \times [A] \rightarrow [A]$	Takes one of the two parties a subtree returns.	
$\text{include} \triangleq (1 \times (\text{list}(\pi_2) \text{ concat})) \text{ cons}$	$\text{include} : F(A, [A] \times [A]) \rightarrow [A]$	The party that invites the root, which puts every immediate subtree's root out. A map.	
$\text{exclude} \triangleq (1 \times (\text{list}(\text{choose}) \text{ concat})) \pi_2$	$\text{exclude} : F(A, [A] \times [A]) \rightarrow [A]$	The party that leaves the root out, so each subtree is free to choose. Not a map.	
$S \triangleq (\text{include}, \text{exclude})$	$S : F(A, [A] \times [A]) \rightarrow [A] \times [A]$	The algebra: one step returns both parties of a subtree at once.	
$\text{party} \triangleq (S) \text{ choose}$	$\text{party} : \text{tree } A \rightarrow [A]$	Every guest list the president's ruling allows.	
the specification $\frac{\text{party}}{\ni} \max R$	$\frac{\text{party}}{\ni} \max R : \text{tree } A \rightarrow [A]$	A guest list of greatest total conviviality.	



node	the algebra's input $A \times [[A] \times [A]]$	include	exclude
d, e, f	(d, [])	[d]	[]
b	(b, [[(d),[]], ([e],[])])	[b]	[d,e], [d], [e], []
c	(c, [[(f),[]])	[c]	[f], []
a	(a, [[(b),...), ([c],...)])	[a] # exclude(b) # exclude(c)	choose(b) # choose(c)

(17.4e)

with the subscripted ratings, $\max R$ keeps ($[b] = 2$, $[d,e] = 9$) at **b** and ($[c] = 3$, $[f] = 6$) at **c**, so at the root $\text{include} = [a,d,e,f] = 16$ beats $\text{exclude} = [d,e,f] = 15$, and choose takes 16.

$$\boxed{\frac{\text{party}}{\ni} \max R} \xrightarrow{\text{definition of party}} \boxed{\frac{\langle S \rangle \text{choose}}{\ni} \max R} \xrightarrow{(9b), \text{absorption}} \boxed{\frac{\langle S \rangle}{\ni} E(\text{choose}) \max R} \quad (17.4f)$$

$$\text{Ex 7.38, } (R^\circ \times R^\circ) \sqsupseteq \text{choose} \sqsubseteq \text{choose } R^\circ \quad \boxed{\frac{\langle S \rangle}{\ni} \max(R \times R) \frac{\text{choose}}{\ni} \max R}$$

$$(17.3e), (1 \times \text{list}(R^\circ \times R^\circ)) \sqsupseteq S \sqsubseteq S (R^\circ \times R^\circ) \quad \boxed{\langle \frac{S}{\ni} \max(R \times R) \rangle \frac{\text{choose}}{\ni} \max R}$$

$$\text{Ex 7.15, the fork splits} \quad \boxed{\langle \langle \frac{\text{include}}{\ni} \max R, \frac{\text{exclude}}{\ni} \max R \rangle \rangle \frac{\text{choose}}{\ni} \max R}$$

$$\text{include a map, } \max \sqsupseteq R \text{ into each branch} \quad \boxed{\langle \langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{\ni} \max R) \text{ concat} \rangle \rangle \frac{\text{choose}}{\ni} \max R}$$

the program is $\text{party} = \langle S \rangle \frac{\text{choose}}{\ni} \max R$ with exclude renamed to $\pi_2 \text{ list}(\frac{\text{choose}}{\ni} \max R) \text{ concat}$ — include was already a map, and exclude becomes one.

§17.5. Shortest paths on a cylinder

definition	type	note	(17.5a)
$F(A, X) = A + A \times X$	$F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.	
$L = \text{list}^+$	$L : \mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.	
α the initial algebra	$\alpha : F(A, L A) \rightarrow L A$	$\alpha = [\text{wrap}, \text{cons}]$: a path is started by one square, or extended by one.	
N the n-tuple relator	$N : \mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.	
$R \triangleq \text{sum} \leq \text{sum}^\circ$	$R : L \text{ Nat} \rightarrow L \text{ Nat}$	The cost of a path, which the cheapest minimises.	
setify	setify : $N A \rightarrow E A$	Forgets which row a component came from, so the n answers become one set.	
moves	moves : $N A \rightarrow E(N A)$	The three columns a path may step to: rotated up, unrotated, rotated down.	
trans	trans : $E(N A) \rightarrow N(E A)$	Transposes a set of tuples, so every row collects its own paths.	
zip	zip : $F(N A, N B) \rightarrow N F(A, B)$	Commutates N with the base functor, pairing each new square with its own row's paths.	
$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$\text{cp} : F(A, E B) \rightarrow E(F(A, B))$	Turns a square paired with a set of paths into the set of extensions of those paths.	
$\text{generate} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$	$\text{generate} : F(N A, N(E(L A))) \rightarrow N(E(L A))$	One fold step: extends every path of every row by the new column.	
$\text{paths} \triangleq (\text{generate}) \text{ setify union}$	$\text{paths} : L N \text{ Nat} \rightarrow E(L \text{ Nat})$	Every path across the cylinder.	
the specification $\text{paths min } R$	$\text{paths min } R : L N \text{ Nat} \rightarrow L \text{ Nat}$	A cheapest path from the entry side to the exit side.	

$$\begin{array}{c}
 \boxed{\text{paths min } R} \xlongequal{\text{definition of paths}} \boxed{(\text{generate}) \text{ setify union min } R} \xrightarrow{(17.1b) \text{ at } \text{union}, R \text{ a preorder}} \boxed{(\text{generate}) \text{ setify } P(\text{min } R) \text{ min } R} \quad (17.5b) \\
 \\
 \boxed{\text{setify}} \xrightarrow{\text{lax natural}} \boxed{(\text{generate}) N(\text{min } R) \text{ setify min } R} \xrightarrow{(14.4a) \text{ at } Q} \boxed{Q \text{ setify min } R} \\
 \\
 \boxed{\text{generate } N(\text{min } R)} \xrightarrow{(7.13), \text{ then zip, trans, moves lax natural}} \boxed{F(1, N(\text{min } R)) Q} \\
 \\
 \boxed{Q} \xlongequal{\text{the fusion condition read as a definition}} \boxed{F(1, \text{moves trans } N(\text{min } R)) \text{ zip } N(\alpha)} \\
 \\
 \boxed{\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]} \xlongequal{\quad} \boxed{[N(\text{wrap}), (1 \times \text{moves trans } N(\text{min } R)) \text{ zip}' N(\text{cons})]}
 \end{array}$$

(7.13) is $F(1, \text{min } R) \alpha \sqsubseteq \text{cp } P(\alpha) \text{ min } R$, Theorem 7.1 at the map α with $\frac{F(1, \exists)}{\exists} \alpha = \text{cp } P(\alpha)$: extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.

§17.6. The security van problem

(17.6a)

definition	type	note
$R \triangleq \text{length} \leq \text{length}^\circ$	$R : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The order the schedule is minimised over: fewer van visits is better.
$\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \max \leq$	$\text{ceiling} : [\text{Int}] \rightarrow \text{Int}$	The highest the bank's balance reaches over a stretch of transactions.
$\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \min \leq$	$\text{floor} : [\text{Int}] \rightarrow \text{Int}$	The lowest it reaches, so $\text{ceiling} - \text{floor}$ is the cash the stretch has to carry.
secure the coreflexive on x with $\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$	$\text{secure} : [\text{Int}] \rightarrow [\text{Int}]$	The stretches one van visit can serve: some starting reserve keeps the cash between 0 and N .
ok the coreflexive on (a, xs) with xs non-empty and $[a] \# \text{head } xs$ secure	$\text{ok} : \text{Int} \times [[\text{Int}]] \rightarrow \text{Int} \times [[\text{Int}]]$	The test the final program runs, in place of old's secure .
$\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$	$\text{new} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	Calls the van: the transaction opens a segment of its own.
$\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$	$\text{glue} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	Puts the transaction on the front of the first segment.
$\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons } \text{secure}) \times 1) \text{ cons}$	$\text{old} : \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	glue restricted to a first segment that stays secure.
$\text{partition} = ([\text{nil}, \text{new} \cup \text{glue}])$	$\text{partition} : [\text{Int}] \rightarrow [[\text{Int}]]$	Every splitting of the transactions into consecutive segments.
$S \triangleq [\text{nil}, \text{new} \cup \text{old}]$	$S : 1 + \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$	The algebra whose fold is every splitting of the transactions into secure segments.
$\text{partition list}(\text{secure}) = (S)$	$\text{partition list}(\text{secure}) : [\text{Int}] \rightarrow [[\text{Int}]]$	Fusion: keeping only secure segments is what turns glue into old .
$H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$	$H : [[\text{Int}]] \rightarrow [[\text{Int}]]$	One schedule's first segment is a prefix of the other's, or both are empty.
$R ; H \triangleq R \cap (R^\circ \Rightarrow H)$	$R ; H : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The refinement of R that makes both halves of S monotonic.
$ R \triangleq R \cap \neg R^\circ$	$ R : [[\text{Int}]] \rightarrow [[\text{Int}]]$	The strict part R splits into: $R ; H = R \cup (R \cap H)$.
the specification $\frac{\text{partition list}(\text{secure})}{\ni} \min R$	$\frac{\text{partition list}(\text{secure})}{\ni} \min R : [\text{Int}] \rightarrow [[\text{Int}]]$	A schedule with the fewest secure segments.

(17.6b)

$$\boxed{\frac{\text{partition list}(\text{secure})}{\ni} \min R} \quad (14.4a), \text{secure prefix} \sqsubseteq \text{prefix} \quad \boxed{\frac{(S)}{\ni} \min R} \quad (7.14) \text{ holds, } (7.15) \text{ FALSE; } R ; H \sqsubseteq R \quad \boxed{\frac{(S)}{\ni} \min(R ; H)}$$

(17.3e), S monotonic on $\overline{R} ; H$ by (7.16) and (7.17) $\boxed{(\frac{S}{\sup} \min(R ; H))}$ old $\sqsubseteq$ new $\overline{(R ; H)}^\circ$ $\boxed{([nil, (ok \rightarrow glue, new)])}$

$\boxed{(1 \times (R ; H)) \text{ old}}$ $R ; H = |R| \cup (R \cap H)$,
 $\cup$ distributes $\boxed{(1 \times |R|) \text{ old} \cup (1 \times (R \cap H)) \text{ old}}$ (7.19) and (7.20) on $|R|$,
 (7.21) on $R \cap H$ $\boxed{\text{new } (R \cap H) \cup \text{old } (R \cap H)}$

$X \cap Y \sqsubseteq X \frac{\sqsubseteq}{\sup} Y$, converses $\boxed{(\text{new} \cup \text{old}) (R ; H)}$

(7.15) $(1 \times R) \text{ old} \sqsubseteq (\text{new} \cup \text{old}) R$ is FALSE — the shorter partition need not stay secure — and that is what refining R to $R ; H$ buys; the second chain is (7.17), and (7.16) rests the same way on (7.18) $(1 \times T) \text{ new} \sqsubseteq \text{new } H$.

§18. Thinning Algorithms

§18.1. Thinning

For $Q : A \rightarrow A$, $\text{thin } Q \triangleq (\exists/\exists) \cap (\exists \backslash (Q^\circ \exists)) : E A \rightarrow E A$ (8.1).

$$ys \text{ (thin } Q) \text{ } xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$$

(18.1a)

the law	what it says	(18.1b)
$X \subseteq \frac{S}{\exists} \text{ thin } Q \iff X \exists \subseteq S \text{ and } S^\circ X \subseteq Q^\circ \exists$	the universal property: everything kept is an S -value, and every S -value has a Q -lower bound among the kept ones	
$Q \subseteq R \implies \text{thin } Q \subseteq \text{thin } R$	the fewer pairs Q relates, the fewer subsets count as thinnings	
$1 \subseteq \text{thin } Q$, and $\text{thin } Q$ is a preorder if Q is	keeping everything is always a legal thinning	
$\min R = \text{thin } Q \min R$ $Q \subseteq R$, both preorders	thin-introduction : thinning first cannot lose an R -minimum	
$\text{thin } Q \supseteq \min Q \frac{1}{\exists}$ (8.2)	thin-elimination : keeping one element is a thinning, but its domain is the sets $\min Q$ is defined on	
$\frac{S}{\exists} \text{ thin } Q \supseteq \frac{S}{\exists} \min R \frac{1}{\exists}$ (8.3), $R \cap (S^\circ S) \subseteq Q$	the usable variant: R need only refine Q between values S gives one argument	
$\text{union thin } Q \supseteq P(\text{thin } Q) \text{ union}$ (8.4)	thinning each member set is a thinning of the union	
$\langle \frac{F(\exists) S}{\exists} \text{ thin } Q \rangle \subseteq \langle \frac{S}{\exists} \rangle \text{ thin } Q$ Theorem 8.1, S monotonic on Q	the thinning theorem : thinning at every step beats thinning only at the end	
$\langle \frac{F(\exists) S}{\exists} \text{ thin } Q \rangle \min R \subseteq \langle \frac{S}{\exists} \rangle \min R$ Corollary 8.1, $Q \subseteq R$ as well	the same against the optimisation problem itself, by thin-introduction	

§18.2. Paths in a layered network

$F(A, X) = A + A \times X$, $L = \text{list}^+$ with initial algebra $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, L A) \rightarrow L A$.

$\text{wrapz} \triangleq \langle \text{wrap}, \text{zero} \rangle$, $\text{consw } (a, (xs, n)) = \langle \text{cons } (a, xs), \text{wt } (a, \text{head } xs) + n \rangle$.

$\text{cost} \triangleq \langle [\text{wrapz}, \text{consw}] \rangle \pi_2$, $\langle [\text{wrapz}, \text{consw}] \rangle = \langle 1, \text{cost} \rangle$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

$Q \triangleq R \cap (\text{head head}^\circ)$, $S \triangleq F(1, \exists) \alpha$, $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$, $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$, $\text{step} \triangleq \text{cpr } P(\text{cons}) \min R$.

(18.2a)

the law	what it says	(18.2b)
$\langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R = \langle \frac{L(\exists)}{\exists} \rangle \min R$	the specification: a least-cost path across the network	
$F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha \ Q$	S is monotonic on Q ; on R it is not, since the next edge can cost arbitrarily much	
$\langle \frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q \rangle \min R \subseteq \langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R$ Corollary 8.1	thinning applies: one partial path per starting vertex is enough	
$S \text{ head} \subseteq [1, \pi_2]$ $S \text{ head}$ simple, so $R \cap (S^\circ S) \subseteq Q$	the side condition of (8.3): between two paths S builds from one argument, equal cost and equal head already means Q	
$\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q$ $\supseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \min R)$ (8.4), (8.3), $P(\frac{1}{\exists}) \text{ union} = 1$	thin eliminated: split the algebra at $F(\exists, 1) F(1, \exists)$, thin each part, one minimum per part	
$\frac{F(1, \exists)}{\exists} P(\alpha) \min R = [\text{wrap}, \text{step}]$	that inner part read off the two branches of α	
$\langle \frac{F(\exists, 1) \alpha}{\exists} \rangle \min R \supseteq \langle [P(\text{wrap}), \text{cpl } P(\text{step})] \rangle \min R$	the program: one fold, a best path per vertex of the current layer	

§18.3. Implementing thin

(18.3a)

$\text{setify} : [A] \rightarrow E\ A$, $\text{cup} : E\ A \times E\ A \rightarrow E\ A$, $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$, $\text{listcp}(F) : F([A]) \rightarrow [F\ A]$, $\text{sort}\ P \triangleq \text{setify}^\circ \text{ordered}\ P$ for P a connected preorder.

$\text{thinlist}\ Q$ is any $\text{thinlist}\ Q \sqsubseteq \text{subseq}$ with $\text{thinlist}\ Q \text{ setify} \sqsubseteq \text{setify thin}\ Q$; one is $\langle [\text{nil}, \text{bump}\ Q] \rangle$, $\text{bump}\ Q\ (a, []) = [a]$, $\text{bump}\ Q\ (a, [b] \# xs) = (b\ Q\ a \rightarrow [a] \# xs, a\ Q\ b \rightarrow [b] \# xs, [a] \# [b] \# xs)$.

Binary thinning data: $S = (f_1\ p_1) \cup (f_2\ p_2)$ with p_1, p_2 coreflexive; Q a preorder with $Q \sqsubseteq R$ and both $f_1\ p_1, f_2\ p_2$ monotonic on Q ; P a connected preorder with both f_1, f_2 monotonic on P ; $g_i \triangleq \text{list}(f_i)\ \text{filter}(p_i)$.

the law	what it says	(18.3b)
$\text{thinlist}\ Q\ xs = [\text{minlist}\ Q\ xs]$ (8.5), Q connected, xs non-empty	what thinning should come to when it can: one element	
$\text{sort}\ P\ \text{thinlist}\ Q \sqsubseteq \text{thin}\ Q\ \text{sort}\ P$ (8.6)	thinning a sorted list is a thinning of the set — this is what $\text{thinlist}\ Q \sqsubseteq \text{subseq}$ buys	
$\text{sort}\ P\ \text{minlist}\ Q \sqsubseteq \text{min}\ Q$ (8.7)	a minimum of the sorted list is a minimum of the set	
$\text{sort}(f\ P\ f^\circ)\ \text{list}(f) \sqsubseteq P(f)\ \text{sort}\ P$ (8.8)	shunt a function through a sort	
$\text{sort}\ P\ \text{filter}(p) \sqsubseteq E(p)\ \text{sort}\ P$ (8.9), p coreflexive	filtering a sorted list sorts the restricted set	
$(\text{sort}\ P \times \text{sort}\ P)\ \text{merge}\ P \sqsubseteq \text{cup}\ \text{sort}\ P$ (8.10)	merging two sorted lists sorts their union	
$F(\text{sort}\ P)\ \text{listcp}(F) \sqsubseteq \text{cp}(F)\ \text{sort}(F\ P)$ (8.11), F linear	$\text{listcp}(F)$ is the list implementation of the cartesian product $\text{cp}(F)$	
$F(\text{sort}\ P)\ \text{listcp}(F)\ \text{list}(f)\ \text{filter}(p) \sqsubseteq \frac{F(\exists)\ f\ p}{\exists}\ \text{sort}\ P$ Lemma 8.1, f monotonic on P , p coreflexive	one sorted list built from sorted arguments, instead of a set built and then sorted	
$\langle \text{listcp}(F)\ (g_1, g_2)\ \text{merge}\ P\ \text{thinlist}\ Q \rangle \text{minlist}\ R$ $\sqsubseteq \frac{\langle S \rangle}{\exists}\ \text{min}\ R$ Theorem 8.2	the binary thinning theorem : a fold on sorted lists of partial solutions, thinned at every step	

§18.4. The knapsack problem

(18.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$, $\text{value} \triangleq \text{list}(\text{vol})\ \text{sum}$, $\text{weight} \triangleq \text{list}(\text{wt})\ \text{sum}$.

$\text{subseq} = \langle [\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2] \rangle$, $\text{within}\ w$ the coreflexive on xs with $\text{weight}\ xs \leq w$.

$\geq \triangleq \leq^\circ$, $R \triangleq \text{value} \geq \text{value}^\circ$, $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$, $P \triangleq R$.

$F\ A = 1 + \text{Item} \times A$, $\text{listcp}(F) = \text{wrap} + \text{cpr}$, $g_1 \triangleq \text{list}([\text{nil}, \text{cons}])\ \text{filter}(\text{within}\ w) = [\text{list}(\text{nil}), h_1]$, $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$.

$h_1 \triangleq \text{list}(\text{cons})\ \text{filter}(\text{within}\ w)$, $h_2 \triangleq \text{list}(\pi_2)$.

the law	what it says	(18.4b)
$\frac{\text{subseq}\ (\text{within}\ w)}{\exists}\ \text{min}\ R$	the specification: a selection of greatest total value that stays within the capacity w	
$\text{subseq}\ (\text{within}\ w) = \langle ([\text{nil}, \text{cons}]\ (\text{within}\ w)) \cup [\text{nil}, \pi_2] \rangle$ fusion, weights non-negative	the selections that fit are themselves a fold, and the fold's algebra already has binary thinning's shape	
$(1 \times R)\ (\text{cons}\ (\text{within}\ w)) \sqsubseteq \text{cons}\ (\text{within}\ w)\ R$ FALSE	not monotonic on R : a selection of greater value need not still fit once one more item goes in	
$(1 \times Q)\ (\text{cons}\ (\text{within}\ w)) \sqsubseteq \text{cons}\ (\text{within}\ w)\ Q$ $(1 \times Q)\ \pi_2 \sqsubseteq \pi_2\ Q$	both halves are monotonic on Q once ties in value are broken by weight	
$\langle \text{listcp}(F)\ (g_1, g_2)\ \text{merge}\ R\ \text{thinlist}\ Q \rangle \text{minlist}\ R$ Theorem 8.2 at $P \triangleq R$, F linear	binary thinning, sorting in descending order of value	
$\text{knapsack}\ w = \langle [\text{nil}, \text{cpr}\ (h_1, h_2)\ \text{merge}\ R\ \text{thinlist}\ Q] \rangle \text{minlist}\ R$	the program; $\text{minlist}\ R$ becomes head , since packings come out in descending value	

§18.5. The paragraph problem

(18.5a)

$\text{Line} = \text{list}^+ \text{ Word}, \text{ Para} = \text{list}^+ \text{ Line}, F A = \text{Word} + \text{Word} \times A, \text{listcp}(F) = \text{wrap} + \text{cpr}.$
 $\text{new } (a, xs) = [[a]] \# xs, \text{glue } (a, xs) = [[a] \# \text{head } xs] \# \text{tail } xs, \text{partition} \triangleq \lambda [wrap \text{ wrap}, \text{new } u \text{ glue}] : \text{list}^+ \text{ Word} \rightarrow \text{Para}.$
 $\text{width} \triangleq \lambda [length, (length \times 1) \text{ plus } succ], \text{fits } w \text{ the coreflexive on a line } x \text{ with width } x \leq w, \text{ok } w \text{ the coreflexive on } [x] \# xs \text{ with width } x \leq w.$
 $\text{white } w \ x = w - \text{width } x, \text{collect} \triangleq \text{list}(\text{sqr } \text{sum}), \text{waste } w \triangleq \text{init list}(\text{white } w) \text{ collect}.$
 $R \triangleq (\text{waste } w) \leq (\text{waste } w)^\circ, Q \triangleq R \cap (\text{head } \text{head}^\circ), P \triangleq T.$
 $g_1 \triangleq \text{list}([wrap \text{ wrap}, \text{new}]), g_2 \triangleq \text{list}([wrap \text{ wrap}, \text{glue}]) \text{ filter}(\text{ok } w), \text{start} \triangleq \text{wrap } \text{wrap } \text{wrap}, h_1 \triangleq \text{list}(\text{new}), h_2 \triangleq \text{list}(\text{glue}) \text{ filter}(\text{ok } w).$

the law	what it says
$\frac{\text{partition list}^+(\text{fits } w)}{\ni} \min R$	the specification: least waste among the paragraphs whose every line fits
$\text{partition list}^+(\text{fits } w) = \lambda [wrap \text{ wrap}, \text{new } u \text{ glue } (\text{ok } w)]$ fusion, every word fits on a line by itself	the fitting paragraphs are themselves a fold
$[wrap \text{ wrap}, \text{new } u \text{ glue } (\text{ok } w)]$ $= [wrap \text{ wrap}, \text{new}] \cup ([wrap \text{ wrap}, \text{glue}] (\text{ok } w))$	rewritten into binary thinning's shape $(f_1 \ p_1) \cup (f_2 \ p_2)$
$(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$	glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this
$(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$ $(1 \times Q) (\text{glue } (\text{ok } w)) \sqsubseteq \text{glue } (\text{ok } w) Q$ cons monotonic on collect $\leq \text{collect}^\circ$	both halves are monotonic on Q once ties in waste are broken by the first line
merge $T = \text{cat}; P \triangleq \text{head prefix head}^\circ$ also serves	T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input
$\lambda [\text{listcp}(F) \ (g_1, g_2) \text{ cat thinlist } Q] \text{ minlist } R$ Theorem 8.2 at $P \triangleq T$	binary thinning, one candidate kept per first line
paragraph $w = \lambda [\text{start}, \text{cpr } (h_1, h_2) \text{ cat thinlist } Q] \text{ minlist } R$	the program, g_1 and g_2 split along the coproduct

(18.5b)

§18.6. Bitonic tours

(18.6a)

$F A = (\text{City} \times \text{City}) + (\text{City} \times A)$, the base functor of cons-lists of length at least two; $\text{listcp}(F) = \text{wrap} + \text{cpr};$
 $\text{tc} : \text{City} \times \text{City} \rightarrow \text{Real}$, neither positive nor symmetric.
 $\text{start } (a, b) = ([a, b], [a, b]), \text{dropl } (a, ([b] \# xs, ys)) = ([a] \# xs, [a] \# ys), \text{dropr } (a, (xs, [b] \# ys)) = ([a] \# xs, [a] \# ys), \text{tour} \triangleq \lambda [\text{start}, \text{dropl } u \text{ dropr}].$
 $\text{cost } (xs, ys) = \text{outcost } xs + \text{incost } ys, \text{outcost } [a_0, \dots, a_n] = \text{tc } (a_0, a_1) + \dots + \text{tc } (a_{n-1}, a_n), \text{incost } [a_0, \dots, a_n] = \text{tc } (a_1, a_0) + \dots + \text{tc } (a_n, a_{n-1}).$
 $\text{next} \triangleq \text{tail head}, \text{next2} \triangleq \text{next} \times \text{next}, R \triangleq \text{cost} \leq \text{cost}^\circ, Q \triangleq R \cap (\text{next2 } \text{next2}^\circ), P \triangleq T, g_1 \triangleq \text{list}([\text{start}, \text{dropl}]), g_2 \triangleq \text{list}([\text{start}, \text{dropr}]).$

the law	what it says	(18.6b)
$\frac{\text{tour}}{\ni} \min R$	the specification: a least-cost bitonic tour, outward journey and return kept as a pair of lists	
$(1 \times R) \text{ dropl} \sqsubseteq \text{dropl } R \quad \text{FALSE}$ $(1 \times R) \text{ dropr} \sqsubseteq \text{dropr } R \quad \text{FALSE}$	neither drop is monotonic on R : the two edges it adds and removes depend on head and next of both lists	
$(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl } Q$ $(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr } Q$	both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input	
$(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \text{ minlist } R$ Theorem 8.2 at $P \triangleq T$, merge $T = \text{cat}$	binary thinning, one candidate per pair of second cities	
$\text{mintour} = ([\text{start wrap, cpr } (\text{list}(\text{dropl}), \text{list}(\text{dropr})) \text{ cat thinlist } Q]) \text{ minlist } R$	the program: quadratic, because each step adds just two tours to the list kept	

§19. Dynamic Programming

§19.1. Theory

$h : F B \rightarrow B$ a map, $T : F A \rightarrow A$ an F -algebra, $R : B \rightarrow B$.

(19.1a)

$H \triangleq (T)^\circ (h) : A \rightarrow B$, $M \triangleq \frac{H}{\supset} \min R$ the problem to be solved, $(\mu X : G(X))$ the least fixed point of G .

the law	what it says	(19.1b)
$(\mu X : \frac{T^\circ}{\supset} P(F(X) h) \min R) \sqsubseteq M$ Theorem 9.1, h monotonic on R	dynamic programming: decompose in every way, solve each part, keep one optimum per part	
$\frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq M$ (9.1)	all Knaster–Tarski leaves to prove	
$\frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq H$ (9.2) $H^\circ \frac{T^\circ}{\supset} P(F(M) h) \min R \sqsubseteq R$ (9.3)	(9.1) split by the universal property of $\min$	
$P(X) \min R \sqsubseteq (\exists X) \cap (\exists \backslash (X R^\circ))$ (9.4) = (7.10), (17.1b)	the only fact about $\min$ either proof uses	
$(\mu X : \frac{T^\circ}{\supset} \text{thin } Q P(F(X) h) \min R) \sqsubseteq M$ Theorem 9.2, Q a preorder with $Q F(H) h \sqsubseteq F(H) h R$; $\text{thin } Q$ as in (18.1b)	the same with a thinning step: decompositions that can never win are dropped	
the fixed point is unique, and entire Theorem 6.3, T° followed by F 's membership relation inductive; $\frac{T^\circ}{\supset}$ finite and non-empty, R connected	when the recursion can be refined to a recursive function	
$\frac{[V_1, V_2]^\circ}{\supset} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \min R$ = $(\text{Ran } V_1 \rightarrow W_1, W_2)$ $W_i \triangleq \frac{V_i^\circ}{\supset} \text{thin}(Q_i) P(U_i) \min R$ Proposition 9.1, $V_2 \frac{V_1^\circ}{\supset} = \perp$	$F A$ is usually a coproduct, and disjoint ranges split the fixed point into one branch per summand	
$F(R) h \sqsubseteq h R$ $Q F(H) h \sqsubseteq F(H) h R$	the two conditions to be checked, in the order the three propositions below discharge them	
$F(R) h \sqsubseteq h R$ Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\text{cost}) k$, $F(\leq) k \sqsubseteq k \leq$	monotonicity when the cost is itself a fold with a step k monotonic on $\leq$	
$F(R \cap (H^\circ H)) h \sqsubseteq h R$ Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$, $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle) k$, $F(\leq \times \mathbb{1}) k \sqsubseteq k \leq$, H° simple	monotonicity in context : k may also read the input the part was built from	
$Q \triangleq F(U, V)$ Proposition 9.4, U, V preorders, $F(U, R) h \sqsubseteq h R$, $V H \sqsubseteq H R$	both conditions of Theorem 9.2 at once, split along the two arguments of a bifunctor	

§19.2. The string edit problem

$Op ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}$, $F(A, B) = 1 + (A \times B)$, $\alpha \triangleq [\text{nil}, \text{cons}]$.

(19.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [Op] \rightarrow [\text{Char}] \times [\text{Char}]$, base returning $([], [])$, $\text{step}(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$, $\text{step}(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$, $\text{step}(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$.

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{succ}])$, $R \triangleq \text{length} \leq \text{length}^\circ$, $U \triangleq \tau$, $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$, $Q \triangleq \mathbb{1} + (U \times V)$, empty the coreflexive on (xs, ys) with both lists empty.

unstep implements $\frac{\text{step}^\circ}{\supset} \text{thin}(U \times V)$: $\text{unstep}([a] \# xs, []) = [(\text{del } a, (xs, []))]$, $\text{unstep}([], [b] \# ys) = [(\text{ins } b, ([], ys))]$, $\text{unstep}([a] \# xs, [b] \# ys) = (a = b \rightarrow [(\text{cpy } a, (xs, ys))], [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$.

the law	what it says	(19.2b)
$\frac{\text{edit}^\circ}{\supset} \min R$	the specification: a shortest edit sequence from which both strings can be reconstituted	
$F(R) \alpha \sqsubseteq \alpha R$ Proposition 9.2 at <code>length</code> , <code>succ</code> monotonic on $\leq$	<code>cons</code> is monotonic on R , so Theorem 9.1 already applies	
$Q F(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$	the thinning condition sought, over $F(0p, [\text{Char}] \times [\text{Char}])$	
$F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise	$U \triangleq T$ costs nothing: any two operations may be compared	
$\text{edit} (1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$ $\text{edit} (\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$	the other half of Proposition 9.4 for $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$	
$\text{edit} (1 \times \text{tail}) \sqsubseteq R^\circ \text{edit}$ $\text{edit} (\text{tail} \times 1) \sqsubseteq R^\circ \text{edit}$ $\text{suffix} = \text{tail}^*$, and $B A \sqsubseteq C B \implies B A^* \sqsubseteq C^* B$	one step is enough: drop the operation that produced the head, or weaken its <code>cpy</code> to a <code>del</code> — never lengthening the sequence	
$(\mu X : \frac{[\text{base}, \text{step}]^\circ}{\supset} \text{thin } Q P([\text{nil}, (1 \times X) \text{cons}]) \min R) \sqsubseteq \frac{\text{edit}^\circ}{\supset} \min R$ Theorem 9.2	a copy, when available, beats a delete or an insert	
$X = (\text{empty} \rightarrow \text{nil}, \frac{\text{step}^\circ}{\supset} \text{thin}(U \times V) P((1 \times X) \text{cons}) \min R)$ Proposition 9.1	<code>base</code> and <code>step</code> have disjoint ranges	
$\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{cons}) \text{minlist } R)$	the program: at most two decompositions survive, but the same subproblem is solved many times, so the running time is exponential	
$\text{column } xs \text{ } ys = [\text{mle } (u, ys) \mid u \leftarrow \text{tails } xs]$ $\text{column } xs = ([\text{fstcol } xs, \text{nextcol } xs]), \text{fstcol} = \text{list}(\text{del}) \text{ tails}$	the tabulation: $\text{mle } (xs, ys)$ needs $\text{mle } (u, v)$ for every tail u of xs and v of ys , so the columns are built right to left	
$\text{column } xs ([b] \# ys) = \text{nextcol } xs (b, \text{column } xs \text{ } ys)$ $\text{nextcol } xs (b, us) = ([[\text{base } (b, \text{last } us), \text{step } b]] \# xus)$ $xus = \text{zip } (xs, \text{zip } (\text{init } us, \text{tail } us))$	each column is a fold built bottom to top, over xs zipped with the adjacent pairs of the column to its right	
$\text{base } (b, u) = [[\text{ins } b] \# u]$ $\text{step } b ((a, (u, v)), ws) =$ $(a = b \rightarrow [[\text{cpy } a] \# v] \# ws,$ $[\text{bmin } R ([\text{del } a] \# w, [\text{ins } b] \# u)] \# ws)$ $w = \text{head } ws$; these <code>base</code> , <code>step</code> are not <code>edit</code> 's	an entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths	

§19.3. Optimal bracketing

<code>tree A ::= tip A bin (tree A, tree A), F X = A + X², so F(R) = 1 + R²; h $\triangleq$ [tip, bin], flatten $\triangleq$ ([wrap, cat]) : tree A $\rightarrow$ list⁺ A, H = flatten^o.</code> (cost, size) $\triangleq$ ([opt, opb]), opt $\triangleq$ (zero, st), opb ((cx, sx), (cy, sy)) = (cb (sx, sy) + cx + cy, sb (sx, sy)). sb associative, so size = flatten sz for a map sz; R $\triangleq$ cost $\leq$ cost^o, g $\triangleq$ [zero, (1 $\times$ sz)² opb π_1], single the coreflexive on singleton lists. splits $\triangleq$ (inits⁺, tails⁺) zip, an implementation of $\frac{\text{cat}^\circ}{\supset}$; array $\triangleq$ inits list(row), row $\triangleq$ tails list(mct), col $\triangleq$ inits list(mct). mix $\triangleq$ zip list(bin) minlist R, next $\triangleq$ (π_1, mix) snoc, process $\triangleq$ ((tip wrap) $\times$ 1) loop(next).	(19.3a)
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the law	what it says	(19.3b)
$\frac{\text{flatten}^\circ}{\ni} \min R$	the specification: a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$, the tree whose flattening is the given list	
$[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) g \quad (9.5)$	the cost of a node reads only the cost and the flattening of its two subtrees	
$(1 + (\leq \times 1)^2) g \sqsubseteq g \leq \quad (9.6)$	g is monotonic on $\leq$ in its two cost arguments	
$F(R \cap (\text{flatten} \text{ flatten}^\circ)) h \sqsubseteq h R$ Proposition 9.3, $H^\circ = \text{flatten}$ a map	monotonicity in context: only trees with the same flattening are compared	
$(\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\ni} P([\text{tip}, (X \times X) \text{ bin}]) \min R) \sqsubseteq \frac{\text{flatten}^\circ}{\ni} \min R$ Theorem 9.1	split the list in every way, bracket both halves, join	
$X = (\text{single} \rightarrow \text{wrap}^\circ \text{ tip}, \frac{\text{cat}^\circ}{\ni} P((X \times X) \text{ bin}) \min R)$ Proposition 9.1	wrap and cat have disjoint ranges	
$\text{mct} = (\text{single} \rightarrow \text{head tip}, \text{splits list}((\text{mct} \times \text{mct}) \text{ bin}) \text{ minlist } R)$	the program; exponential, since the segments of one list overlap	
$\text{mct} = (\text{single} \rightarrow \text{head tip}, (\text{init col}, \text{tail row}) \text{ mix}) \quad (9.7)$	the tabulation: $\text{mct } xs$ is needed for every non-empty segment xs , so the values are held as an array of rows	
$\text{col} = (\text{single} \rightarrow \text{head tip wrap}, (\text{init col}, \text{tail row}) \text{ next}) \quad (9.8)$ $\text{cons col} = (1 \times \text{array}) \text{ process} \quad (9.9)$ $\text{row} = (\text{single} \rightarrow \text{head tip wrap}, (\text{mct}, \text{tail row}) \text{ cons}) \quad (9.10)$	a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop	
$\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip wrap wrap}$ $\text{addcol} \triangleq (\pi_1 \text{ tip wrap}, \text{step}) \text{ cons}$ $\text{step} \triangleq (\text{process tail}, \pi_2) \text{ zip list}(\text{cons})$	the program: one fold building the array column by column, cubic in the length of the input	

§19.4. Data compression

$[A] ::= \text{nil} \mid \text{snoc } ([A], A), \text{ list}^+ A ::= \text{wrap } A \mid \text{snoc } (\text{list}^+ A, A), \text{ String} = [\text{Char}], \text{ Code} ::= \text{sym Char} \mid \text{ptr } (\text{String}, \text{String}^+), F(\text{Code}, \text{String}) = 1 + (\text{String} \times \text{Code}), \alpha \triangleq [\text{nil}, \text{snoc}].$ $\text{decode} \triangleq ([\text{nil}, \text{extend}]) : [\text{Code}] \rightarrow \text{String}, \text{extend } (xs, \text{sym } a) = xs \# [a], \text{extend } (xs, \text{ptr } (ys, zs)) = xs \# zs \text{ when } ys \# zs \text{ is a proper prefix of } xs \# zs; H = \text{decode}^\circ.$ $\text{size} \triangleq ([\text{zero}, \text{distr } [1 \times c, 1 \times p] \text{ plus}])$ with c, p the constant costs of a symbol and a pointer; $R \triangleq \text{size} \leq \text{size}^\circ, Q \triangleq F(\tau + \tau, \text{prefix}^\circ) = 1 + (\text{prefix}^\circ \times (\tau + \tau))$, the two τ on symbols and on pointers. $\text{lrt } ws = \min(\text{prefix}^\circ \times (\tau + \tau)) \{(xs, (ys, zs)) \mid xs \# zs = ws, ys \# zs \text{ a proper prefix of } ws\}$, the longest repeated tail; $\text{reduce } (ws \# [a]) = (zs \neq [] \rightarrow [(ws, \text{sym } a), (xs, \text{ptr } (ys, zs))], [(ws, \text{sym } a)])$ with $(xs, (ys, zs)) = \text{lrt } (ws \# [a]).$	(19.4a)
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the law	what it says
$\frac{\text{decode}^\circ}{\ni} \min R$	the specification: a smallest code sequence decoding to the given string
$F(R) \alpha \sqsubseteq \alpha R \quad (\tau + \tau) [c, p] = [c, p]$	monotonicity is routine, the two costs being constants
$F(\tau + \tau, R) \alpha \sqsubseteq \alpha R$ $\text{decode prefix} \sqsubseteq R^\circ \text{ decode}$ Proposition 9.4 at $Q \triangleq F(\tau + \tau, \text{prefix}^\circ)$	the thinning condition split in two: pointers are ordered only by how much input they consume, symbols and pointers not at all
$\text{decode init} \sqsubseteq R^\circ \text{ decode}$	enough for the second: drop the last character of the output by shortening or removing the last code element, never raising the cost
$(\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{\ni} \text{thin } Q P([\text{nil}, (X \times 1) \text{snoc}]) \min R) \sqsubseteq \frac{\text{decode}^\circ}{\ni} \min R$ Theorem 9.2	between a symbol and a pointer nothing can be decided in advance; between two pointers the longer match wins
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau)) P((X \times 1) \text{snoc}) \min R)$ Proposition 9.1	nil and extend have disjoint ranges
$\frac{\text{extend}^\circ}{\ni} (ws \# [a]) = \{(ws, \text{sym } a)\} \cup \{(xs, \text{ptr } (ys, zs)) \mid xs \# zs = ws \# [a], ys \# zs \text{ a proper prefix of } ws\}$	the decompositions of one string: take the last character as a symbol, or end with a pointer
$\text{reduce implements } \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau + \tau))$	thinning leaves at most two: the symbol, and the pointer of lrt
$\text{encode} = (\text{null} \rightarrow \text{nil}, \text{reduce list}((\text{encode} \times 1) \text{snoc}) \text{minlist } R)$	the program, again exponential; the book gives no tabulation for it

§20. Greedy Algorithms

§20.1. Theory

h, T, R, H, M as in (19.1a); additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\supseteq}$ returns, so that $\frac{T^\circ}{\supseteq} \min Q$ is entire. (20.1a)

the law	what it says
$(\mu X : \frac{T^\circ}{\supseteq} \min Q F(X) h) \sqsubseteq M$ Theorem 10.1, the hypotheses of Theorem 9.2	greedy : one decomposition is kept at every step, so P and $\frac{T^\circ}{\supseteq}$ disappear from the recursion (20.1b)
$\frac{[V_1, V_2]^\circ}{\supseteq} \min(Q_1 + Q_2) [U_1, U_2]$ $= (Ran V_1 \rightarrow W_1, W_2)$ $W_i \triangleq \frac{V_i^\circ}{\supseteq} \min(Q_i) U_i$ Proposition 10.1, $V_2 V_1^\circ = \perp$	Proposition 9.1 with min for thin
$Q \triangleq F(U, V)$ Proposition 9.4	still the way to get both conditions, but $\frac{T^\circ}{\supseteq} \min Q$ must now be entire as well
Theorem 7.2, (17.3e)	that greedy theorem chooses among the results of one relational step of a reduce; Theorem 10.1 chooses among the decompositions of the input, for an arbitrary T rather than an initial algebra, at the cost of h being a map

§20.2. The detab-entab problem

$detab \triangleq ([nil, expand]) : String \rightarrow String$ over snoc-lists, $\alpha \triangleq [nil, snoc]$, $H = detab^\circ$; $expand (xs, a) = (a = TB \rightarrow fill\ xs, xs \# [a])$, $fill\ xs = xs \# blanks\ (n - (col\ xs) \bmod n)$. (20.2a)

$col \triangleq ([zero, count])$, $count (c, a) = (a = NL \rightarrow 0, c + 1)$; TB the tab, BL the blank, NL the newline, tab stops every n columns.

$R \triangleq length \leq length^\circ$, U the preorder with $a U b \iff a = TB \vee a = b$, $V \triangleq prefix^\circ n (fill\ fill^\circ)$, $Q \triangleq 1 + (V \times U)$.

$unfill\ xs$ the shortest prefix of xs with $fill\ (unfill\ xs) = fill\ xs$; tbc the trailing blank count, $triple \triangleq (unfill\ entab, (tbc, col))$.

the law	what it says
$\frac{\text{detab}^\circ}{\supset} \min R$	the specification: a shortest input detab expands to the given output — detab entab = 1 and nothing shorter does
$F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise	U may be any preorder on characters; $a U b \iff a = TB \vee a = b$ puts TB below every character, so min prefers a tab to a blank
$\text{detab prefix} \sqsubseteq R^\circ \text{detab}$ FALSE	at $n = 8$, detab [a,b,c,d,e,TB] = [a,b,c,d,e,BL,BL,BL], whose prefix [a,b,c,d,e,BL,BL] is longer than any input giving it
$\text{detab } V^\circ \sqsubseteq R^\circ \text{detab} \quad V \triangleq \text{prefix}^\circ n \text{ (fill fill}^\circ)$	true once only the prefixes that do not cross a tab stop are allowed
$\text{nil } V^\circ = \text{nil} \quad \text{fill } V^\circ = \text{fill}$ $\text{snoc } V^\circ \sqsubseteq \text{snoc } U \text{ (}\pi_1 V^\circ\text{)}$ left as exercises	the three properties of V the derivation rests on; the second is what fill fill [°] was added for
$\text{expand } V^\circ \sqsubseteq \text{expand } U \text{ (}\pi_1 V^\circ\text{)}$	the claim they add up to: shortening the output either leaves the last step alone or discards it
$\text{detab } V^\circ \sqsubseteq \text{detab } U \text{ (init detab } V^\circ)$ <i>init</i> inductive, so the greatest solution is the unique one	hence $\text{detab } V^\circ \sqsubseteq \text{prefix detab}$, and prefix $\sqsubseteq R^\circ$ finishes Proposition 9.4
$(\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{\supset} \min Q (1 + (X \times 1)) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{\supset} \min R$ Theorem 10.1	greedy: one character of input is decided at each step
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{expand}^\circ}{\supset} \min(V \times U) (X \times 1) \text{snoc})$ Proposition 10.1	nil and expand have disjoint ranges
$\frac{\text{expand}^\circ}{\supset} (xs \# [a]) = \{(ys, TB) \mid \text{fill } ys = xs \# [a]\} \cup \{(xs, a)\}$ the first set is non-empty iff $a = BL$ and $\text{col } (xs \# [a]) \bmod n = 0$	the last output character came from a tab only if it is a blank landing exactly on a tab stop
$\frac{\text{expand}^\circ}{\supset} \min(V \times U) (xs \# [a]) = (a = BL \wedge \text{col } (xs \# [a]) \bmod n = 0 \rightarrow (\text{unfill } xs, TB), (xs, a))$	the greedy step: emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop
$\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (\text{tbc } xs) \text{ (10.1)}$	what makes triple a snoc -list reduce: the output splits at the last tab stop
$\text{triple} = ([\text{base}, \text{op}])$ and $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$	the program: one pass carrying the column and the count of pending blanks
base returns $([], (0, 0))$ op $((xs, (t, c)), a) =$ $(a = BL \wedge (c + 1) \bmod n \neq 0 \rightarrow (xs, (t + 1, c + 1)),$ $a = BL \rightarrow (xs \# [TB], (0, c + 1)),$ $a = NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)),$ $(xs \# \text{blanks } t \# [a], (0, c + 1)))$	hold a blank back, cash the held blanks in for a tab at a tab stop, flush them at a newline, flush them before anything else

§20.3. The minimum tardiness problem

$F X = 1 + (X \times \text{Job})$, $\alpha \triangleq [\text{nil}, \text{snoc}]$ on schedules, $\beta \triangleq [\text{nil}, \text{snag}]$ on bags, **snag** putting a job into a bag; **bagify** $\triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}$, $H = \text{bagify}^\circ$.

ct, **dt**, **wt** : **Job** $\rightarrow$ **Real** the completion, due and weighting quantities of a job; **penalty** $(xs, j) = (\text{sum } (\text{list}(\text{ct}) xs) + \text{ct } j - \text{dt } j) \times \text{wt } j$.

cost $\triangleq \frac{\text{prefix}}{\supset} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{max} \leq$, **cost** $[] = 0$, **cost** $(xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (xs, j))$, $R \triangleq \text{cost} \leq \text{cost}^\circ$.

perm $\triangleq \text{bagify bagify}^\circ = ([\text{nil}, \text{add}])$, **add** $(xs, j) = ys \# [j] \# zs$ for some $xs = ys \# zs$.

$k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{penalty})) \text{bmax}]$, $f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]$, $Q \triangleq f \leq f^\circ$.

the law	what it says	(20.3b)
$\frac{\text{bagify}^\circ}{\ni} \min R$	the specification: an ordering of the given bag of jobs with least maximum penalty	
$F(R \cap (\text{bagify } \text{bagify}^\circ)) \alpha \sqsubseteq \alpha R \quad (10.2)$	monotonicity in context — only schedules of one bag are compared	
$(Q \cap (\beta \beta^\circ)) F(\text{bagify}^\circ) \alpha \sqsubseteq F(\text{bagify}^\circ) \alpha R \quad (10.3)$	the greedy condition, also in context	
$\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (\text{perm } xs, j))$ $\alpha \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k, F(\leq \times 1) k \sqsubseteq k \leq$ Proposition 9.3	(10.2): penalty sums the completion times of xs , and a sum does not see the order	
$\alpha \text{ cost} = \langle g, h \rangle \text{bmax} \quad (10.4)$ $g \triangleq [\text{zero}, \text{penalty}] \quad (10.5)$ $h \triangleq [\text{zero}, \pi_1 \text{ cost}] \quad (10.6)$	the cost of a schedule split into the penalty of its last job and the cost of the rest	
$\text{add} \sqsubseteq \pi_1 R \quad (10.7)$ $\beta \text{ bagify}^\circ = F(\text{bagify}^\circ) [\text{nil}, \text{add}] \quad (10.8)$	adding a job never lowers the cost; and bagify° is itself a reduce on bags. Together: $\beta \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) h \leq \text{cost}^\circ$	
$F(\text{bagify}) Q^\circ F(\text{bagify}^\circ) = g \geq g^\circ$, met by $Q \triangleq f \leq f^\circ, f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]$	the specification Q has to satisfy, and the choice that meets it: a Q -minimum is a job of least penalty	
no greedy reduce exists	a greedy snoc-list reduce would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole	
$X = (\text{null} \rightarrow \text{nil}, \frac{\text{snag}^\circ}{\ni} \min Q (X \times 1) \text{snoc}) \quad \text{Theorem 10.1, Proposition 10.1}$	nil and snag have disjoint ranges	
$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick } (\text{schedule} \times 1) \text{snoc})$ $\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\ni} \min Q$, a partial function	the program: repeatedly remove a job of least penalty and put it last; quadratic in the number of jobs	

§20.4. The TeX problem

$\text{intern} \triangleq \text{val round} : \text{Decimal} \rightarrow [0, 2^{16}), \text{val} \triangleq ()[\text{zero}, \text{shift}], \text{shift } (d, r) = (d + r)/10, \text{round } r \text{ rounds } 2^{16} r \text{ to the nearest integer: } \text{round } r = n \iff 2n - 1 < 2^{17} r < 2n + 1.$ $\text{interval } n = ((2n - 1)/2^{17}, (2n + 1)/2^{17}), r \text{ inrange } (a, b) \iff a < r < b, \text{round}^\circ = \text{interval inrange}, R \triangleq \text{length} \leq \text{length}^\circ.$ Interval the pairs (a, b) with $0 < b < 1$ and $a < b$ (10.9) $[\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}$, $\text{step } (d, (a, b)) = ((d + a)/10, (d + b)/10).$ $F X = 1 + (\text{Digit} \times X), \alpha \triangleq [\text{nil}, \text{cons}], h \triangleq ()[\text{arb}, \text{step}] = H^\circ, ! : \text{Digit} \times \text{Interval} \rightarrow 1, Q \triangleq (\text{!}^\circ \text{!}^\circ \text{!}_r) \cup 1, w \triangleq 2^{17}.$	(20.4a)
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the law	what it says
$\frac{\text{intern}^\circ}{\supset} \min R = \text{interval} \frac{\text{inrange val}^\circ}{\supset} \min R$	the specification: a shortest decimal whose internal representation is the given multiple of 2^{-16} . round° is not a map, but interval is, so it comes out of the $\frac{\supset}{\supset}$
$\text{val inrange}^\circ = ([\text{arb}, \text{step}])$ fusion; $\text{zero inrange}^\circ = \text{arb}, \text{shift inrange}^\circ = (1 \times \text{inrange}^\circ) \text{ step}$	the converse of val , cut down to intervals, is a reduce on cons-lists
$F(R) \alpha \sqsubseteq \alpha R$	cons is monotonic on R — routine
$\alpha^\circ F(h) Q^\circ \sqsubseteq R^\circ \alpha^\circ F(h)$	the greedy condition of Theorem 10.1, converted
$\frac{\text{arb}^\circ}{\supset} (a, b) = (a < 0 \rightarrow \{*\}, \{\})$ $\frac{\text{step}^\circ}{\supset} (a, b) = \{(d, (10a - d, 10b - d)) \mid 0 < 10b - d < 1\}$	the decompositions of an interval: stop, or take one more digit
$0 < 10b - d_1 < 1 \text{ and } 0 < 10b - d_2 < 1 \text{ imply } d_1 = d_2$	step° is in fact a map, d the digit with $0 < 10b - d < 1$, so $\frac{[\text{arb}, \text{step}]^\circ}{\supset}$ returns at most two elements and Q need only choose between them
$Q \triangleq (\iota_l^\circ !^\circ \iota_r) \cup \mathbb{1}$, and $! \text{ nil} \sqsubseteq \text{cons } R^\circ$ from $! \text{ nil length} \sqsubseteq \text{cons length} \geq$	stop whenever stopping is legal: the empty decimal is shorter than any other
$(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\supset} \min Q F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\supset} \min R$ Theorem 10.1	greedy: emit the one digit the interval allows, until the interval contains zero
$\text{extern} = \text{interval } f$ $f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a - d, 10b - d))$	the program, with d the digit above
$\text{extern } n = f(2n - 1, 2n + 1)$ $f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p - w d, 10q - w d))$ $d = (10q) \text{ div } w, w = 2^{17}$	the same in integer arithmetic only, as chapter 3 required of intern : every interval reached is $(p/w, q/w)$

§21. Appendix

§21.1. $P(S) \min R = (\exists S) \cap (\epsilon \backslash (S R^\circ))$

(p, q) tabulates $W \triangleq (\exists S) \cap (\epsilon \backslash (S R^\circ))$, and $y \triangleq \frac{(p \exists S) \cap (q R)}{\exists}$, a map.

$$\begin{array}{ccc} \boxed{\frac{W}{P(S) \min R}} & \xleftarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} & \begin{array}{c} \boxed{\frac{p^\circ y}{P(S)}} \\ \boxed{\frac{y^\circ q}{\min R}} \end{array} \end{array} \quad (21.1a)$$

$$\begin{array}{ccc} \boxed{\frac{y^\circ q}{\exists}} & \xleftarrow{f^\circ \cdot \neg f \cdot, \cdot \exists \neg \exists} & \boxed{\frac{q}{(p \exists S) \cap (q R)}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\frac{p^\circ q}{\exists S}} \\ \boxed{\frac{1}{R}} \end{array} \\ & & \text{R reflexive} \end{array} \quad (21.1b)$$

$$\boxed{\frac{\epsilon y^\circ q}{R^\circ}} \xleftarrow{\cdot \exists \neg \exists, \circ, \text{meet}} \boxed{\frac{R^\circ q^\circ q}{R^\circ}} \quad \text{q simple} \quad (21.1c)$$

$$\begin{array}{ccc} \boxed{\frac{p^\circ y}{P(S)}} & \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, \circ, f^\circ \cdot \neg f \cdot} & \begin{array}{c} \boxed{\frac{p^\circ y \exists}{\exists S}} \\ \boxed{\frac{p \exists}{y \exists S^\circ}} \end{array} \end{array} \quad (21.1d)$$

$$\boxed{\frac{p^\circ y \exists}{\exists S}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\frac{p^\circ p \exists S}{\exists S}} \quad \text{p simple}$$

$$\boxed{\frac{p \exists}{y \exists S^\circ}} \xleftarrow{\text{modular law}} \boxed{\frac{p \exists}{(p \exists) \cap (q R S^\circ)}} \xleftarrow{1 \sqsubseteq q q^\circ} \boxed{\frac{q^\circ p \exists}{R S^\circ}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\frac{W}{\epsilon \backslash (S R^\circ)}} \quad \text{W's right half}$$

§21.2. $P(\min R) \min R = P(\text{Dom}(\min R)) \text{ union } \min R$

$$\boxed{P(\min R) \min R} \xrightarrow{\text{Dom}(\min R) \min R = \min R} \boxed{P(\text{Dom}(\min R) \min R) \min R} \quad (21.2a)$$

$$\xrightarrow{P \text{ a relator}} \boxed{P(\text{Dom}(\min R)) P(\min R) \min R} \xrightarrow{(17.1.9a)} \boxed{P(\text{Dom}(\min R)) \text{ union } \min R} \quad (21.2b)$$

$$\boxed{\frac{P(\text{Dom}(\min R)) \text{ union } \min R}{P(\min R) \min R}} \xleftarrow{(21.1b) \text{ at } S := \min R, \Delta \neg \cap, T \cdot \neg T \setminus} \begin{array}{c} \boxed{\frac{P(\text{Dom}(\min R)) \text{ union } \min R}{\exists \min R}} \\ \boxed{\frac{\epsilon P(\text{Dom}(\min R)) \text{ union } \min R}{\min R R^\circ}} \end{array}$$

$$\boxed{\frac{P(\text{Dom}(\min R)) \text{ union } \min R}{\exists \min R}} \xleftarrow{P(\text{Dom}(\min R)) \sqsubseteq 1} \boxed{\frac{\text{union } \min R}{\exists \min R}}$$

$$\xleftarrow{T(U \cap V) \sqsubseteq T U \cap T V, \cdot \exists \neg \exists, (17.1.4a)} \boxed{\frac{(\exists \exists) \cap (\epsilon \backslash (\epsilon \backslash R^\circ))}{\exists \min R}} \xleftarrow{\text{modular law}} \boxed{\frac{\exists (\exists \cap (\epsilon \backslash (\epsilon \backslash R^\circ)))}{\exists (\exists \cap (\epsilon \backslash R^\circ))}} \quad \text{counit of } T \cdot \neg T \setminus$$

$$\boxed{\frac{\epsilon P(\text{Dom}(\min R)) \text{ union } \min R}{\min R R^\circ}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\frac{\text{Dom}(\min R) \epsilon \text{ union } \min R}{\min R R^\circ}}$$

$$\begin{array}{ccc}
 \overleftarrow{\overline{\text{Dom}(\min R) \sqsubseteq \min R (\min R)^\circ}} & \boxed{\begin{array}{c} \min R (\min R)^\circ \in \text{union } \min R \\ \min R R^\circ \end{array}} & \overleftarrow{\overline{(\min R)^\circ \sqsubseteq \epsilon, \epsilon \in \text{union } \sqsubseteq \epsilon}} \quad \boxed{\begin{array}{c} \min R \in \min R \\ \min R R^\circ \end{array}} \\
 & & \text{UP of min}
 \end{array}$$